

Estimation of atmospheric methane emissions between 1996 and 2001 using a three-dimensional global chemical transport model

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[1] Using an atmospheric inversion approach, we estimate methane surface emissions for different methane regional sources between 1996 and 2001. Data from 13 high-frequency and 79 low-frequency CH₄ observing sites have been averaged into monthly mean values with associated errors arising from instrumental precision, mismatch error, and sampling frequency. Simulated methane mole fractions are generated using the 3-D global chemical transport model (MATCH), driven by NCEP analyzed observed meteorology (T62 resolution), which accounts for the impact of synoptic and interannually varying transport on methane observations. We adapted the Kalman filter to optimally estimate methane flux magnitudes and uncertainties from seven seasonally varying (monthly varying flux) and two aseasonal sources (constant flux). We further tested the sensitivity of the inversion to different observing sites, filtered versus unfiltered observations, different model sampling strategies, and alternative emitting regions. Over the 1996–2001 period the inversion reduces energy emissions and increases rice and biomass burning emissions relative to the a priori emissions. The global seasonal emission peak is shifted from August to July because of increased rice and wetland emissions from southeast Asia. The inversion also attributes the large 1998 increase in atmospheric CH₄ to global wetland emissions. The current CH₄ observational network can significantly constrain northern emitting regions but not tropical emitting regions. Better estimates of global OH fluctuations are also necessary to fully describe interannual methane observations. This is evident in the inability of the optimized emissions to fully reproduce the observations at Samoa.

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1. Introduction

[2] Methane is the second most radiatively important greenhouse gas attributable to human activity. It contributes about 15% of the total 2.5 W m^{-2} increase in radiative forcing caused by the anthropogenic release of greenhouse gases in the industrial age [Hansen and Sato, 2001]. Direct, worldwide observations over the past two decades show the atmospheric burden of methane increasing at approximately 0.5% per year. Sources of methane arise from both natural (e.g., wetlands, termites) and anthropogenic (e.g., rice, domesticated animals, natural gas) processes, but large uncertainties exist in their individual magnitudes and temporal variabilities on the global and regional scale. A better

understanding of the current methane budget is necessary to predict possible changes and feedbacks due to climate change, and to sensibly formulate emission reduction strategies.

[3] Estimation of surface fluxes of methane and many other trace gases on the global scale have relied on three broad approaches. The first approach is extrapolation of few direct flux (or proxy flux) measurements to larger regions (e.g., Matthews and Fung [1987] for wetlands and Lerner *et al.* [1988] for animals). The second approach uses process models that represent the actual physical and biological processes of methane production (e.g., Walter *et al.* [2001a] and Cao *et al.* [1996b] for wetlands). These first and second methods are considered to be “bottom-up” approaches, and suffer from extrapolation and process model errors, respectively. This study uses a third method known as the “top-down” or atmospheric inverse modeling approach. Here, worldwide methane observations and simulations from atmospheric chemical transport models are combined to estimate unknown CH₄ fluxes and their uncertainties. This approach has errors arising from insufficient observations and model errors.

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[4] Most previous methane inversion studies have used weekly flask measurements, a single year of meteorological transport, and solved for multiyear or seasonally averaged methane fluxes or flux trends [e.g., *Fung et al.*, 1991; *Hein et al.*, 1997; *Houweling et al.*, 1999; *Dentener et al.*, 2003; *Cunnold et al.*, 2002]. Over the past two decades, however, high-frequency in situ monitoring stations that measure atmospheric methane have become active (e.g., Advanced Global Atmospheric Gases Experiment (AGAGE)). These high-frequency measurements capture the full variability of CH₄ observations, including the occurrence of subweekly synoptic transport events [*Chen and Prinn*, 2005]. They offer, in principle, much greater information about methane sources, sinks, and transport compared to weekly measurements. In addition to high data frequency, realistic driving meteorology (as opposed to climatological or a single year of winds) is required to accurately determine of interannual and intra-annual fluxes. This is because interannual transport can strongly affect the tracer observations. *Chen and Prinn* [2005] showed the strong year-to-year influence of El Niño and the North Atlantic Oscillation (NAO) on methane mole fractions at Samoa and Mace Head, respectively, from transport alone. *Warwick et al.* [2002] also examined the global influence of transport IAV on atmospheric CH₄ growth rates.

[5] In this study, methane emissions from different regional sources and their uncertainties are optimally estimated between 1996 and 2001 at monthly time resolution. The inversion incorporates high- and low-frequency observations and realistic meteorology. We use the Model for Atmospheric Transport and Chemistry (MATCH) driven by National Center for Environmental Prediction (NCEP) analyzed observed winds at a resolution ($\sim 1.8^\circ \times 1.8^\circ$) higher than previous global studies. We also adapt the Kalman filter to optimize methane emissions at monthly time resolution. In addition to computational convenience, the Kalman filter quantifies the usefulness of each additional measurement in an observational time series. Optimized interannual and averaged monthly emissions and uncertainties are compared to previous bottom-up and top-down estimates.

[6] We also test the sensitivity of the inversion results to different observational choices and modeling scenarios. In addition to varying the observational networks, we compare the use of unfiltered (i.e., pollution events retained) versus filtered (i.e., pollution events removed) observations. We also compare the use of modeled monthly means using all model time steps within a month versus time steps corresponding to exact observational times. Finally, the potential impact of model errors is discussed. In general, the timescale of the results (e.g., multiyear average vs. monthly anomalies) will determine which model error dominates.

[7] This paper is organized as follows: Methane observations, the MATCH model (including OH sink), and methane sources are described in sections 2, 3, and 4, respectively. The Kalman filter inverse methodology is described in section 5. Section 6 focuses on the optimized seasonal cycle and annually averaged results, and includes observational and model sensitivity tests. Section 7 describes the interannual methane fluxes for seven seasonal

sources at monthly time resolution. The conclusions are contained in section 8.

2. Observations

[8] An initial step in this study was the accumulation, intercalibration, and organization of available methane time series. Table 1 lists the sites names, locations, and laboratory affiliations, as well as error information described later in section 5.1. The site locations are also plotted in Figure 1. Although methane observations exist at other sites, we include only those 92 sites that were active during our 1996 to 2001 period of study, are of reliable quality, and can be adequately represented using a global CTM. As much in situ high-frequency data as possible was included. The 13 high-frequency stations (sites 1–13), listed in Table 1 and shown in Figure 1, measure methane mole fractions in situ between 24 and 36 times per day [e.g., *Prinn et al.*, 2000]. The 74 low-frequency sites (see Table 1 and Figure 1) are locations where flask samples of air are collected, with measurement at a later date [e.g., *Climate Monitoring and Diagnostics Laboratory (CMDL)*, 2001b]. Most sites are located in the Northern Hemisphere; tropical land regions which have important methane sources in particular are undersampled (Figure 1). We have further divided the flask measurements into sites that have more (41 sites) and less (38 sites) than 42 out of 60 (i.e., 70%) of the monthly averaged observations between July 1996 and June 2001, which is the 5-year time period of the inversion. For brevity, Table 1 includes only those 41 flask sites that were 70% active (sites 14–54); the other flask sites are included in the auxiliary material.¹

[9] With 5 active sites, the Advanced Global Atmospheric Gases Experiment (AGAGE) operates the greatest number of high-frequency stations over the longest time period. *Chen and Prinn* [2005] conducted a modeling analysis of the these high-frequency observations using the identical version of MATCH as used here. Peak fluctuations, including pollution events that typically last a few days or less, that are well characterized by high-frequency sampling can be reproduced by MATCH. The high frequency of in situ measurements is a significant advantage over flask sampling, which typically has a temporal resolution of one observation per week. However, flask sampling allows greater spatial coverage because samples need only be collected, rather than measured, at a particular site. The Climate Monitoring and Diagnostics Laboratory (CMDL, NOAA) operates the greatest number of flask sites, in conjunction with several other laboratories. In this study, duplicate and triplicate flask measurements have been averaged to produce single CH₄ mole fractions at each time.

[10] For both in situ and flask observations, we have discarded samples flagged as having obvious contamination, but retained those samples labeled as polluted or nonbaseline (e.g., AGAGE [*Cunnold et al.*, 2002] and CMDL [*Conway et al.*, 1994]). The “Filt” column in Table 1 lists whether or not observations at a particular site include identified pollution events over 1996–2002. A data set with pollution events removed is considered to

¹Auxiliary material is available at <ftp://ftp.agu.org/apend/jd/2005jd006058>.

Table 1. Methane Measuring Site Information for High-Frequency and Flask Observations^a

Site	Name	Location	Latitude	Longitude	Altitude, m	Network Affiliation ^b	Filt.	Flask Error	Sample Frequency Error	Mismatch Error	Total Error
<i>13 High-Frequency Sites</i>											
1	alt	Alert, Greenland	82	−62	210	4	N	0.0	0.8	4.9	6.5
2	brw	Barrow, Alaska	71	−156	11	2	Y	0.0	0.8	21.5	21.9
3	mhd	Mace Head, Ireland	53	−9	25	1	Y	0.0	1.2	14.8	15.4
4	frd	Fraserdale, Canada	49	−81	250	4	N	0.0	2.0	13.9	14.6
5	coi	Cape Ochi-Ishi, Japan	43	145	100	5	N	0.0	1.9	6.8	8.1
6	thd	Trinidad Head, CA	41	−124	140	1	Y	0.0	1.4	8.2	9.3
7	iza	Tenerife, Canary Islands	28	−16	2360	2	N	0.0	2.1	6.0	7.6
8	mnm	Minamitorishima, Japan	24	153	8	6	N	0.0	2.3	5.4	7.2
9	hat	Hateruma	24	123	47	5	N	0.0	3.5	14.5	15.5
10	mlo	Mauna Loa, Hawaii	19	−155	3397	2	Y	0.0	0.6	5.0	6.5
11	rpb	Barbados	13	−59	42	1	Y	0.0	0.4	3.6	5.5
12	smo	Samoa	−14	−170	42	1	N	0.0	0.3	2.5	4.9
13	cgo	Cape Grim, Australia	−41	145	94	1	Y	0.0	0.4	3.1	5.2
<i>54 Flask Sites Operational for at Least 70% of the Months Used in the Inversion</i>											
14	zep	Zeppelin St., Norway	78	11	474	2	Y	1.8	11.7	3.0	13.3
15	stm	Atlantic Ocean, Norway	66	2	7	2	Y	1.4	11.5	4.2	13.2
16	ice	Storhofdi, Iceland	63	−20	100	2	Y	1.6	14.6	4.8	16.2
17	Sis	Shetland Is., Scotland	60	−1	30	3	Y	5.0	26.7	5.9	29.4
18	bal	Baltic Sea, Poland	55	16	7	2	Y	1.5	22.3	19.7	30.1
19	cba	Cold Bay, Alaska	55	−162	25	2	Y	1.5	6.3	3.4	8.7
20	shm	Shemya Island, Alaska	52	174	40	2	Y	1.6	8.8	2.7	10.6
21	epc	Estevan Pt, BC, Canada	49	−126	39	3	Y	1.8	18.2	9.2	21.1
22	hun	Hegyatsal, Hungary	46	16	344	2	Y	1.7	38.2	93.6	101.2
23	lef	Park Falls, Wisconsin	45	−90	868	2	Y	1.8	16.5	13.3	21.9
24	uum	Ulaan Uul, Mongolia	44	111	914	2	Y	1.6	15.0	6.8	17.3
25	kzd	Sary Taukum, Kazakhstan	44	77	412	2	Y	1.1	17.6	18.8	26.1
26	bsc	Black Sea, Romania	44	28	3	2	Y	2.0	24.4	39.0	46.4
27	kzm	Plateau Assy, Kazakhstan	43	77	2519	2	Y	1.4	14.7	16.2	22.4
28	nwr	Niwot Ridge, Colorado	40	−105	3475	2	Y	1.4	11.7	10.3	16.3
29	uta	Wendover, Utah	39	−113	1320	2	Y	1.7	11.6	24.6	27.7
30	azr	Azores	38	−27	40	2	Y	1.8	12.2	3.5	13.8
31	tap	Tae-ahn Pen., Korea	36	126	20	2	Y	2.0	32.3	44.0	54.9
32	wlg	Mt. Wanliguan, China	36	100	3810	2	Y	1.6	14.9	16.0	22.5
33	lmp	Lampedusa, Italy	35	12	85	2	N	5.0	12.8	10.7	19.8
34	bme	Bermuda East	32	−64	30	2	Y	1.7	15.0	7.4	17.5
35	bmw	Bermuda West	32	−64	30	2	Y	1.7	15.5	7.9	18.2
36	wis	Sede Boker, Israel	31	34	400	2	Y	1.3	13.6	10.4	17.8
37	mid	Midway	28	−177	4	2	N	1.4	10.6	4.9	12.7
38	key	Key Biscayne, FL	25	−80	3	2	Y	1.7	25.4	39.0	46.8
39	ask	Assekrem, Algeria	23	5	2728	2	N	1.8	8.5	5.0	11.2
40	kum	Kumukahi, Hawaii	19	−154	3	2	Y	2.1	8.2	3.4	10.6
41	gmi	Guam	13	144	2	2	Y	1.5	6.5	6.0	10.2
42	sey	Mahe Island, Seychelles	−4	55	3	2	N	1.5	8.7	9.9	14.1
43	asc	Ascension Island	−7	−14	54	2	Y	1.6	1.9	1.7	5.8
44	cfa	Cape Ferguson, Aust.	−19	147	2	3	Y	2.3	8.6	3.6	11.2
45	eic	Easter Island	−27	−109	50	2	Y	1.7	1.9	0.5	5.6
46	crz	Crozet, Indian Ocean	−46	51	120	2	Y	1.8	2.4	0.3	5.9
47	mqa	Macquarie Island	−54	158	12	3	Y	2.2	3.8	0.5	7.1
48	tdf	Teirra Del Fuego	−54	−68	20	2	N	1.6	4.7	2.4	7.5
49	psa	Palmer Station, Antarct	−64	−64	10	2	N	1.5	2.1	0.5	5.5
50	maa	Mawson St., Antarctica	−67	62	32	3	Y	2.5	1.8	0.2	6.7
51	syo	Syowa Station, Antarct.	−69	39	11	2	Y	1.8	1.7	0.3	5.7
52	hba	Halley Bay, Antarctica	−75	−26	10	2	N	1.3	1.4	0.3	5.1
53	spoc	South Pole	−89	−24	2810	3	Y	1.8	2.4	0.9	6.0
54	spo	South Pole	−89	−24	2810	2	Y	2.7	3.5	0.9	7.6
Laboratory	Network	Definition	Intercalibration Factor								
1	AGAGE	Advanced Global Atmospheric Gases Experiment	1								
2	CMDL	Climate Monitoring and Diagnostics Laboratory	1.0119								
3	CSIRO	Commonwealth Scientific and Industrial Research Organization (Australia)	1.0119								
4	MSC	Meteorological Service of Canada	1 ^b								

Table 1. (continued)

Laboratory	Network	Definition	Intercalibration Factor
5	NIES	National Institute for Environmental Studies (Japan)	1
6	JMA	Japan Meteorological Agency	1

^aActive for at least 70% of the months between 1996 and 2001. See Figure 1 for locations and section 5.1 for description of error terms. All errors in ppb CH₄.

^bMSC CH₄ standard is within 0.3% of AGAGE.

be more easily modeled by global models, at the expense of loss of near-station flux information. We test the use of both unfiltered (polluted) and filtered data sets in the inversion. Our study uses only actual observations, and not smoothed/interpolated flask data such as from CMDL [2001b], since we use MATCH's capability to simulate observed values at specific times [Chen and Prinn, 2005].

[11] The standards used for the absolute calibration of methane mole fractions differ among most laboratories, although most have been calibrated to either the AGAGE or CMDL scale. We use the AGAGE standard, which is based on the Tohoku gravimetric technique standard as described by Cunnold *et al.* [2002]. The most recent intercalibration factor of 1.0119 [Cunnold *et al.*, 2002] is used to convert other CH₄ measurements based on the CMDL scale, as done by Chen and Prinn [2005]. The reported precision due to random instrumental error of most methane measurements is between 0.07 and 0.2% [Cunnold *et al.*, 2002; CMDL, 2001a], equivalent to a 1–3 ppb error. These absolute calibration and instrumental precision errors are usually small compared to other types of observational error, as described and quantified in section 5.1.

3. Methane Sources

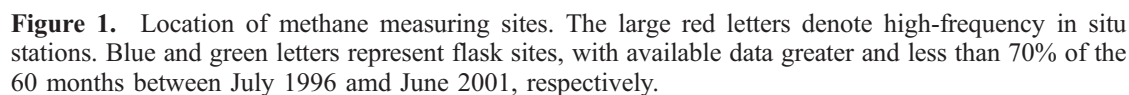
[12] Along with the compilation of the global methane observations, a major preparatory effort in this study was the accumulation of the most up-to-date, readily available surface emission data sets for use in global atmospheric models. A first step was the choice of emission source, since often more than one emission distribution and magnitude is available for a single process. Once chosen, individual emission patterns were then (1) combined to create a “reference” CH₄ flux, representing an initial guess of all methane emissions, and (2) used to define the emission patterns for individual sources whose magnitudes are optimally estimated in the inversion (section 5). For certain processes, this included further spatial aggregation or disaggregation of the original data set.

[13] Surface flux fields of methane were obtained or adapted from data sets described by Fung *et al.* [1991] for wetland and rice, EDGAR 3.0 (Emission Database for Global Atmospheric Research, available at <http://www.mnp.nl/edgar/>) for aseasonal anthropogenic emissions, and Hao and Liu [1994] for biomass burning. Figure 2 shows the individual emission patterns of these data sets at MATCH T62 resolution. The reference emission magnitudes are listed in the first row of Table 2. The original emission magnitudes for the different processes have been scaled to more realistically reflect current bottom-up estimates, with additional uniform scaling of approx-

imately 10% for all emissions to match the CH₄ growth rate, largely determined by the global OH concentration (section 4). Note that these reference emission magnitudes fall within the literature ranges (rows 1 and 2 in Table 2). The relatively high total emission of 589 Tg yr⁻¹ is a result of the prescribed global OH level. A plot representative of the annual average pattern of the total reference emission is given by Chen and Prinn [2005].

[14] The individual sources can also be divided into seasonal (Wetlands, Rice, and Biomass Burning) and aseasonal emissions (AnMSW, and Energy), as shown in Table 2. Seasonal processes, such as wetlands, contain significant variations in magnitude and location in successive months. Aseasonal components, such as emissions from fossil fuel methane leakage, are considered to vary much more slowly, and are assigned a constant spatial distribution over time. We have further divided the total wetland distribution into northeastern (WetNE), northwestern (WetNW), and southern and tropical (WetS) regions. This division allows each of these regions to be solved separately in the inversion, rather than as a whole. Several observational stations surround WetNE and WetNW, allowing the measurement of distinct emission signatures from these two regions. Further disaggregation would be difficult due to the relative sparseness of methane observations within these regions. Note that we have combined wetlands in southeast Asia with rice emission to create a unified emission pattern (RiceWet). These two processes share substantially overlapping spatial and temporal characteristics, especially in southeast Asia, which makes their individual estimation difficult. As shown in Figure 2, greater than 80% of the rice flux originates from China, India, and southeast Asia. In the inversion, the rice region thus represents a large emitting region localized in southeast Asia, with more diffuse emissions spread across other tropical and temperate areas (Figure 2). The partitioning between rice and wetland emissions in RiceWet is 21% wetland (located in Southeast Asia) and 79% rice, based on reference values.

[15] Seasonal ITCZ shifts, which influence tropical precipitation, result in a strong seasonal variation in the distribution of biomass burning in Asia, South America, and Africa [see Hao and Ward., 1993; Hao and Liu, 1994; Duncan *et al.*, 2003b]. This results in a bimodal seasonal behavior of tropical biomass burning emissions. In the inversion, animal and waste (AnMSW) emissions are solved as a combined emission since they have similar spatial patterns. Figure 2 shows their combined distribution, with the greatest emissions near highly populated centers. The Energy emission distribution is a combination of gas and



[16] An additional 60 Tg yr^{-1} source of methane included in the reference run, but not solved for in the inversion, is denoted as “Other” emissions in Table 2. This source includes 36 Tg yr^{-1} from industrial and biofuel combustion sectors (EDGAR 3.0) and 23 Tg yr^{-1} from termites [*Fung et al.*, 1991]. We do not solve for these “other” processes because they relatively diffuse, i.e., distributed over large space scales and therefore measurements are not therefore very sensitive to them. We do not include geologic sources of methane [*Etiopie and Klusman*, 2002; *Milkov et al.*, 2003] and methane hydrates in this study, since their timing, spatial distribution, and emission magnitude are highly uncertain. For the similar reasons, we also ignore methane surface sinks due to bacterial consumption [*Ridgwell et al.*, 1999].

[17] The MATCH model was developed to realistically simulate atmospheric constituents using analyzed observed meteorology [Rasch *et al.*, 1997; Mahowald, 1996]. Throughout this work, MATCH is driven by NCEP reanalysis meteorology at T62 spectral resolution, which corresponds to approximately $1.8^\circ \times 1.8^\circ$ horizontal resolution. In the vertical, NCEP-driven MATCH has 28 sigma levels between ~ 1000 and 2.9 mb. The surface (bottom) layer varies between 50 and 100 m in height. An identical version

[18] This section describes the inversion methodology used to estimate methane emissions and their uncertainties. The methodology is based on the Kalman filter, which has been used in the study of different atmospheric trace gases on the global scale [e.g., *Prinn et al.*, 2001; *Huang*, 2000;

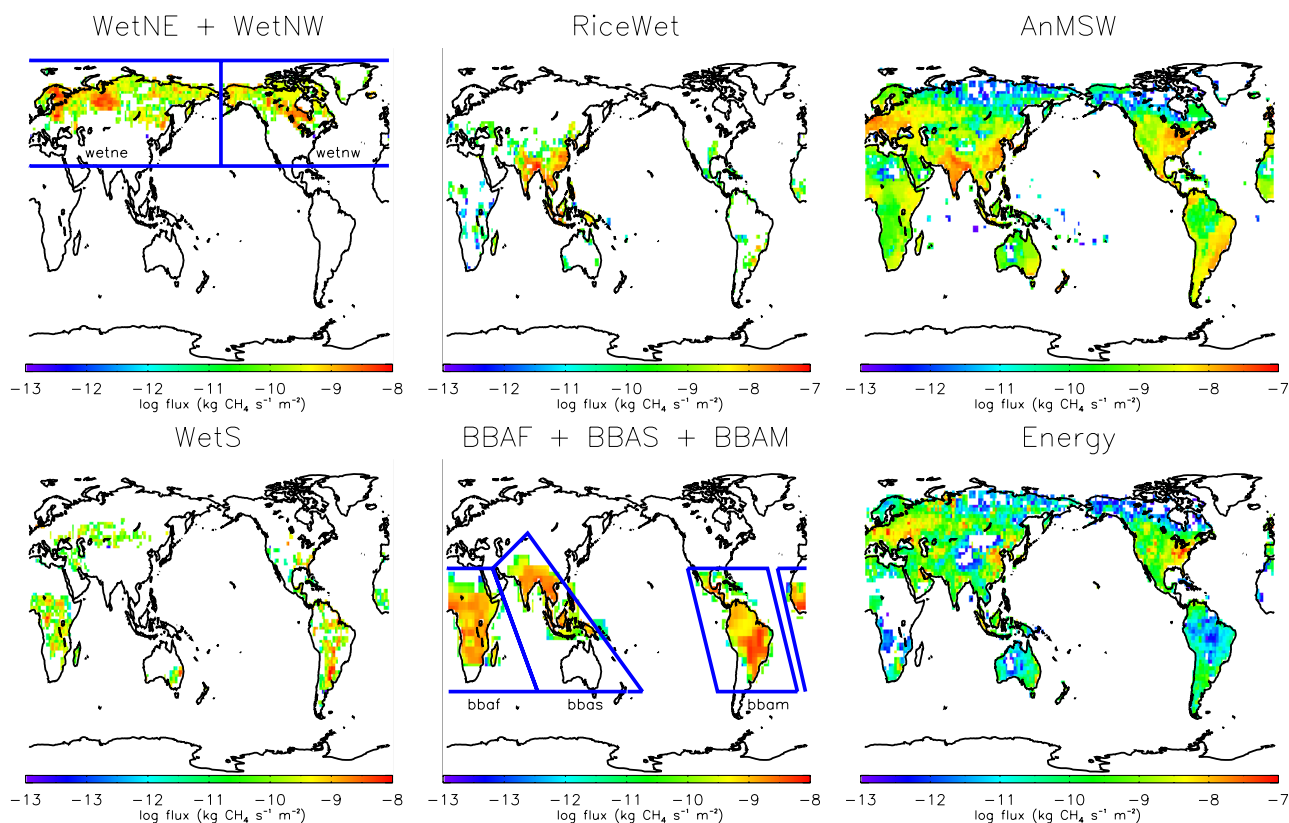


Figure 2. Methane reference source spatial distributions. Annually averaged patterns are shown; note that the seasonal sources (WetNE, WetNW, WetS, RiceWet, BBAF, BBAS, and BBAM) have significant pattern changes from month to month. The aseasonal sources (AnMSW and Energy) have constant emission patterns.

Table 2. Aggregated Methane Sources^a

General Source	Number of Sites	MATCH Interpol.	Filt. Obs.	Wetlands ^b	Rice ^c	Biomass Burning ^d	AnMSW ^e	Energy ^f	Other ^g	Total
Reference total				151 ± 54	92 ± 38	29 ± 19	169 ± 169	89 ± 89	60	589
Literature Range ^h				92–237	25–126	23–55	120–180	53–159		500–600
Inversion cases										
Control	54	Yes	No	145 ± 25	112 ± 29	43 ± 18	189 ± 3	48 ± 3	60	596
HF Only	13	Yes	No	148 ± 33	96 ± 29	37 ± 19	211 ± 4	46 ± 4	60	597
HF MBL ⁱ	14	Yes	No	146 ± 30	110 ± 31	43 ± 18	184 ± 3	54 ± 3	60	596
Control FullAv	92	Yes	No	146 ± 26	111 ± 30	46 ± 18	180 ± 3	54 ± 3	60	596
Control Filt	54	No	No	145 ± 25	114 ± 28	44 ± 18	187 ± 3	48 ± 3	60	597
Control FullAv Filt	92	No	No	145 ± 26	115 ± 29	48 ± 18	175 ± 3	55 ± 3	60	597
All Obs	92	Yes	Yes	143 ± 25	113 ± 29	48 ± 17	185 ± 3	48 ± 3	60	596
All Obs FullAv	54	Yes	Yes	144 ± 25	115 ± 28	47 ± 18	182 ± 3	49 ± 3	60	596
All Obs Filt	54	No	Yes	145 ± 26	115 ± 28	48 ± 18	178 ± 3	52 ± 3	60	597

^aValues are in Tg CH₄ yr⁻¹. The inversion results listed here are a summation of individual regional sources listed in Table 3. Section 5.5 describes the nine inversion cases.

^bWetlands = WetNE + WetNW + WetS + 0.21 × RiceWet (see Table 3).

^cRice = RiceWet × 0.79 (see Table 3).

^dBiomass Burning = BBAF + BBAM + BBAS.

^eAnimal and waste emissions spatial distribution are adapted from EDGAR 3.0.

^fEnergy = Natural Gas + Coal.

^gIncludes 36 Tg yr⁻¹ industrial/biofuel combustion (EDGAR 3.0) and 23 Tg yr⁻¹ termites [Fung et al., 1991].

^hRange from IPCC [2001] except for Wetlands [Walter et al., 2001a] and Energy [Walter et al., 2001a].

ⁱIncludes 21 flask sites that sample marine boundary layer (MBL) air: asc, azr, bme, bmw, cba, chr, crz, eic, hba, ice, kum, mid, psa, shm, spo, syo, tdf, cfa, cgo, maa, mqa.

Mahowald et al., 1997], but not for methane. On the regional scale, the KF has been used to constrain emissions of chlorine compounds [Kleiman and Prinn, 2000] and methane [Janssen et al., 1999]. Here we adapt the Kalman filter to estimate methane emissions from the regional sources described in the previous section at monthly time resolution for seasonally varying emissions and as time-invariant fluxes for more steady emissions. Much of the KF methodology is described by Prinn [2000] and Enting [2002].

[19] Section 5.1 describes the formulation of the observational errors, which determine the relative importance (i.e., weighting) of each observation in the Kalman filter, and consequently, its influence on the final optimized emissions. Sections 5.2–5.4 describe the several equations which make up the full Kalman filter as used in this study. For brevity, we have moved some of the methodological description, including matrix information, to the auxiliary material.

5.1. Observational Errors

[20] The observational error (ϵ_k) associated with the observed (y_k^o) monthly mean at time step k is not known; however, its probability distribution function can be expressed by its standard deviation, σ_k . The overall error in the observed mole fraction, in units of ppb, can be estimated as the net effect of the errors associated with: (1) the instrumental and sampling errors, (2) the sampling frequency used to define the monthly mean, and (3) the mismatch error between observations and model as shown in equation (1). This assumes (reasonably) that these errors are uncorrelated.

$$\sigma_k = \sqrt{\sigma_{\text{measurement}}^2 + \sigma_{\text{mismatch}}^2 + \sigma_{\text{sampling frequency}}^2} \quad (1)$$

The measurement error, $\sigma_{\text{measurement}}$, arises from imperfections in instrumentation, sampling, and intercalibration.

$$\sigma_{\text{measurement}} = \sqrt{\sigma_{\text{instrument precision}}^2 + \sigma_{\text{flask errors}}^2 + \sigma_{\text{intercalibration}}^2} \quad (2)$$

[21] The instrumental precision of CH₄ measurements is approximately 1–4 ppb CH₄ at nearly all sites [Cunnold et al., 2002; CMDL, 2001a]. A uniform and deliberately generous instrumental precision error of 4 ppb is chosen for all sites. After adjusting all measurements to the AGAGE standard (section 2) and assuming perfect accuracy, the calibration error, $\sigma_{\text{intercalibration}}$, should be zero. However, Cunnold et al. [2002] report a 1 ppb difference at a number of colocated AGAGE and CMDL sites even after intercalibration, which we take as $\sigma_{\text{intercalibration}}$. For both in situ and flask measurements, we discard observations that are flagged for obvious contamination or instrumental difficulties. In addition, we retain measurements that are flagged as representing nonbackground (or “polluted”) air, since we generally expect that MATCH can simulate these measurements [see Chen and Prinn, 2005]. However, the inversion sensitivity to filtered (non-polluted) observations is also examined.

[22] The sampling frequency error, $\sigma_{\text{sampling frequency}}$, quantifies how well a monthly mean quantity is defined given a

finite number of (m) measurements. Assuming temporally uncorrelated data, the error on a mean quantity due to limited sampling frequency is best represented as the standard error, following Wunsch [1996]:

$$\sigma_{\text{sampling frequency}} = \sqrt{\frac{\sigma_{\text{mon}}^2}{m}} \quad (3)$$

where σ_{mon}^2 is the monthly mean variance, and m is the number of observations taken in that month. Since σ_{mon}^2 is not known at locations other than the high-frequency stations, it is approximated at most sites using MATCH high-frequency CH₄ output from the reference run. This validity of this assumption is supported by the forward comparisons of Chen and Prinn [2005], which shows similar variability between MATCH modeled and observed high-frequency values. As listed in Table 1, high-frequency observations have a very low monthly standard error given the high number of measurements each month ($m \sim 1000$). The standard error is much higher for the weekly flask measurements ($m \sim 4$). At the same site, the high-frequency standard error would be about 15 to 16 times smaller than the flask error. This quantifies the greater weight of high-frequency measurements in the inversion, especially at locations with a high monthly variance.

[23] The mismatch (or “representation”) error, σ_{mismatch} , describes the difference between an observation made at a single point in space and a model-simulated observation representative of a large volume of air [e.g., Prinn, 2000]. The degree to which a point measurement fails to represent this volume dictates the size of the mismatch error. Among other factors, this difference depends on the resolution of the model, and the observational method and location. Most methane observing sites are situated to sample large, well-mixed volumes of air, which can be more accurately modeled. The mismatch error increases significantly over continental sites near emitting regions, as MATCH does not have the resolution to cope with local influences. The mismatch error is difficult to quantify and may include bias error, where the model systematically overestimates or underestimates the observed mole fractions. Previous studies have used the temporal variability at a particular site as a proxy for this error (e.g., Prinn et al. [2000] for halocarbons at high-frequency stations and Gurney et al. [2002] for CO₂). We have chosen to estimate the mismatch error at each site using the standard deviation of the CH₄ mole fraction y_{ik} (mean $\bar{y}_k$) at the nine model grid cells i which contain and surround each observing site,

$$\sigma_{\text{mismatch}} = \sqrt{\frac{1}{9} \sum_{i=1}^9 (y_{ik} - \bar{y}_k)^2}.$$

This assumes that the spatial variability within a single grid cell is related to the variability among the neighboring grid cells. The mismatch error so defined is much larger over strongly emitting continental sites compared to remote ocean locations. The mismatch error at each site also varies by month, consistent with seasonal changes in emissions and transport.

[24] Table 1 lists individual and total monthly mean errors calculated at each site, averaged between July 1996 and

June 2001. The total error is the aggregate sum of the individual errors, following equation (1). For high-frequency observations, the measurement and mismatch errors have the largest contributions, while the sampling frequency errors are small. In addition to these errors, the flask observations nearby to strongly emitting regions have very large sampling frequency errors. At a given site, the errors on monthly mean observations can vary due to monthly variations in the mismatch and sampling frequency errors. Very large error bars usually correspond to a monthly mean observation defined by only 1 or 2 flask measurements.

5.2. Measurement Equation

[25] The measurement equation relates modeled and observed mole fractions through the following relationship:

$$y_k^o = \tilde{y}_k + H_k x_k + y_{k-T-1}^{adj} + \varepsilon_k \quad (4)$$

The time index k refers to a specific month of observation and flux. The observation vector, y_k^o , contains the monthly mean observations at time k for all sites. $\tilde{y}_k$ contains the corresponding modeled monthly means from the MATCH reference run. The primary objective of the inversion is to obtain estimates of the state vector, x_k , whose individual elements represent adjustments to the reference emissions (section 5.4). These adjustments improve the modeled fit to observed CH₄ monthly means, weighted by observational errors. The elements of x_k are related to the simulated CH₄ mole fractions through the time-dependent sensitivity matrix H_k , which is generated through multiple MATCH runs (described below). Although x_k is synchronized to the monthly observational vector, y_k^o , it also includes elements corresponding to emissions from T months previous to time k , i.e., $k-1$, $k-2$, ..., $k-T$. This is because an observation depends not only on the emission at a particular month, but also on all previous emissions. The contribution of optimized emissions prior to month $k-T$ are contained in y_{k-T-1}^{adj} .

[26] The sensitivity matrix, H , relates the CH₄ mole fraction at each site to the CH₄ emissions in the state vector. Its elements are generated from multiple MATCH runs of the individual methane source of interest (e.g., WetNE, WetNW, etc.) described in section 3. Denoting the reference values with a tilde, the elements of H can be written as:

$$h_{ijkk'} = \frac{y_{ik} - \tilde{y}_{ik}}{x_{jk'} - \tilde{x}_{jk'}} \cong \frac{\partial y_{ik}}{\partial x_{jk'}} \quad (5)$$

In the above, i represents site number and j methane regional source. The sensitivity elements for the aseasonal sources contain the cumulative emission influence of all previous months (month 1 to month k). The seasonal sensitivities only contain emission information for a single specific month (k') prior to the month of interest (k), i.e., $k' \leq k$. The seasonal and aseasonal sensitivities are thus generated by two different types of MATCH simulations. For the aseasonal fluxes (e.g., animal emissions), emissions from a single source are perturbed above their reference levels by 20% in all months and the model then run over the entire data period. The sensitivities are then expressed in units of ppb (Tg yr)⁻¹. For the seasonal sensitivities (e.g., wetland flux from a specific month), a

1-month pulse of methane is emitted and its subsequent dispersion is calculated for all succeeding months.

[27] The Kalman filter, as applied here, requires a linear relationship between sources and mixing ratios. Chen [2003] confirmed the linearity of simulated methane pulses using the same version of MATCH. The use of a prescribed (but optimized) OH field also ignores possible CH₄-OH feedbacks. We expect this effect to be small, however, as the optimized methane adjustments are relatively small and should not impact global OH concentrations substantially. This effect is discussed in the auxiliary material.

5.3. State-Space Equation

[28] This section describes how the state vector (x_k) evolves from one time step to another, before the use of the next monthly observational vector. The state-space equation, equation (6), uses the state transition matrix, M_k to propagate the state vector from one time step to another. In equation (7), this matrix also propagates the state error covariance matrix (P_k) associated with the state vector from one time step to another.

$$x_k^f = M_{k-1} \cdot x_{k-1}^a + \eta_{k-1} \cong M_{k-1} \cdot x_{k-1}^a \quad (6)$$

$$P_k^f = M_{k-1} \cdot P_{k-1}^a \cdot M_{k-1}^T + Q_{k-1} \quad (7)$$

These equations are also known as the “forecast” and “forecast error” equations in other applications. The superscripts f (forecast) and a (analysis) denote a quantity before and after, respectively, the use of one month’s observations to update the state. These equations allow the emissions contained in the state vector to be solved recursively as new observations are added (see section 5.4). A full matrix description of M_k is included in the auxiliary material, but basically shifts the elements of the state vector down. The elements which exit the state vector are considered to be the optimized (adjustment) emissions.

[29] If the state vector included the total seasonal emissions, rather than adjustments to the reference emissions, then M_k would describe the periodic evolution of seasonal emissions from one month to another. An approximation to this seasonal variation is implicitly contained in the reference run, $\tilde{y}_k$. The random forcing term, η_{k-1} , may include multiple sources of error in M_{k-1} . For example, the annually repeating seasonality, implicit in the reference run, cannot describe year-to-year changes in seasonal fluxes at monthly time resolution. This error is not known exactly for each month, but its statistics are contained in Q_{k-1} , the random forcing covariance matrix. This error corresponds to an initial error of the adjustment flux before the use of observations.

5.4. Kalman Filter for Time-Dependent Inversions

[30] The Kalman filter produces an optimal estimate of the state vector with each new monthly observational data set. The previously described state-space equation and measurement equations are combined with the standard Kalman filter gain matrix, state vector update, and error covariance update expressions to yield the full KF equations:

[31] State space extrapolation equations

$$x_k^f = M_{k-1} \cdot x_{k-1}^a \quad (8)$$

$$P_k^f = M_{k-1} \cdot P_{k-1}^a M_{k-1}^T + Q_k \quad (9)$$

Model measurement equation

$$y_k = \tilde{y}_k + H_k x_k^f + y_{k-T-1}^{adj} \quad (10)$$

Kalman gain matrix

$$K_k = \frac{P_k^f H_k^T}{H_k P_k^f H_k^T + R_k} \quad (11)$$

State update

$$x_k^a = x_k^f + K_k \cdot (y_k^o - y_k) \quad (12)$$

State error covariance update

$$P_k^a = (I - K_k H_k) P_k^f \quad (13)$$

These equations are repeated over all monthly observations from $k = 1$ to $k = N$. The measurement equation has been modified to include only modeled CH_4 mole fractions, shown as y_k . The Kalman gain matrix combines the prior state error P_k^f , the observational error R_k (derived in section 5.1), and the sensitivity matrix H_k to compute a weighting matrix. This matrix determines the degree to which observations will modify the prior state through the state update equation, equation (12). A large gain matrix K corresponds to a strong sensitivity of the state vector (emissions) to observations, while a small K corresponds to a very low sensitivity to observations. The optimized state, x_k^a , will almost always undergo some change from the prior state, x_k^f , based on the size of the gain matrix, K_k , and the modeled-observational difference, $y_k^o - y_k$. Equation (13) updates the previous estimated flux uncertainties contained in P_k^f , to the new estimated uncertainty in P_k^a . Note that the diagonal elements of P_k^a are always less than or equal to those contained in P_k^f . This error, or uncertainty, reduction is an indicator of how effectively the observations (given particular observational errors) constrain the fluxes.

[32] Figure 3 shows how the estimates of the WetNE (northwest wetlands) subvector of x corresponding to May 1999 emissions change as observations from subsequent months are used. The starting value for WetNE is zero since it represents an initial adjustments to the reference run emissions. The large initial errors correspond to the diagonal elements of the a priori error covariance for the new emission elements. For most seasonal fluxes the a priori uncertainties (contained in Q in Equation 7) are taken as $\pm 100\%$ of their reference monthly emission. A smaller initial uncertainty of $\pm 30\%$ is used for the much larger emissions from WetS and RiceWet. With each new monthly observation the initial value changes and the error decreases, following equations (8)–(13). Note that the adjustment and error reduction for this source (and all the

other seasonal sources) is greatest in the first 3–4 months, by which time most flux values have stabilized. This stabilization arises from the decreased sensitivity of observations to a given monthly emission due to atmospheric mixing after the first few months. The optimized emission values for May 1999 are obtained after using 11 months of data (i.e., after the March 2000 observation). At each month k , a new subvector of optimized emissions is thus produced. The optimized emissions that exit the state vector are used to update y_{k-T-1}^{adj} , which contains the continued influence of these emissions on the global background methane mole fraction.

[33] The optimization of the aseasonal components are more straightforward. Since constant emissions are assumed, solution toward a single value occurs over all time steps, as shown in Figure 4 for the animal and waste (AnMSW) case. For the aseasonal errors, a large a priori uncertainty equal to $\pm 100\%$ the reference magnitudes was chosen. The rapid decrease in the uncertainty indicates that the inversion is relatively insensitive to this starting value. The optimized flux adjustment is reached with the final observation in the inversion. The optimized total aseasonal emission is the sum of the flux adjustment and the reference value. Because the aseasonal emissions are only fully optimized at the final step of the time series, earlier seasonal emissions (whose estimates also depend on the aseasonal values) at the beginning of the filter will not be fully optimized. In order to fully optimize all seasonal monthly values, the entire Kalman filter is repeated with aseasonal values fixed to their optimized values from the first KF run. The optimized seasonal fluxes show only small changes in this second run.

[34] At each time step, an optimally estimated set of seasonal emissions is produced and collected to create an optimally estimated emission time series over the entire inversion period. Figure 5 (left) shows the monthly emission time series (red) using methane observations between July 1996 and June 2001 for a representative inversion case described in section 5.5. Figure 5 also contains the annually repeating reference emissions (blue). The inversion results contain the average, seasonal, and interannual behavior of methane emissions, which are discussed in sections 6 and 7. The right side of Figure 5 superimposes the optimized (red) uncertainties on top of the reference (blue) uncertainties. Note that the inversion always acts to reduce the initial uncertainty by amounts depending on the sensitivity of the observations to particular source regions, as well as the observational error. The error reduction can be large (e.g., WetNW) or small (e.g., biomass burning regions). For most sources, the final optimized value lies inside the range of the initial uncertainty. Poorly constrained regional fluxes, such as BBAM, may for some months lie slightly outside the initial error bar. The initial and final uncertainties overlap in nearly all cases, however.

[35] A test of the inversion is that the model-observational comparison improves when the using the optimized emissions, compared to the reference. The optimized monthly mean CH_4 mole fractions (red) using the Control case inversion (described in section 5.5) are shown in Figure 6, which also includes the observations (black) and reference run (blue). The errors on the monthly mean observations are shown as vertical bars, as derived in this section 5.1. The

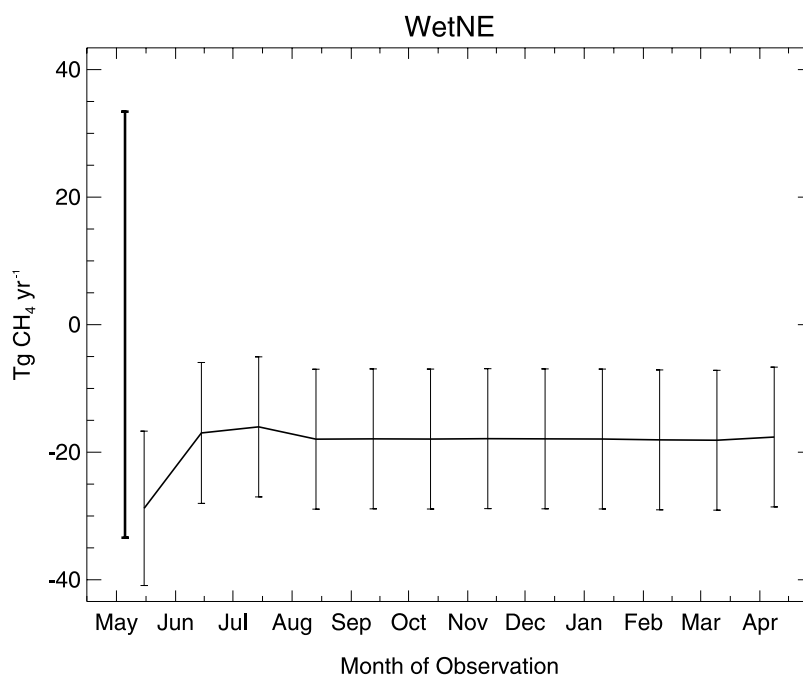


Figure 3. Evolution of estimated WetNE source for a single monthly emission (May 1997). The vertical axis corresponds to emission adjustments from the reference value ($\text{Tg CH}_4/\text{month}$), and the horizontal axis corresponds to the addition of new observational data. The initial vertical line shows the assumed a priori error for the May, 1997 emission. The emission adjustment and error reduction occurs with the use of the first few months of data. The final optimized emission estimate is taken after 11 months of observations have been used, by which time the emission estimate has stabilized.

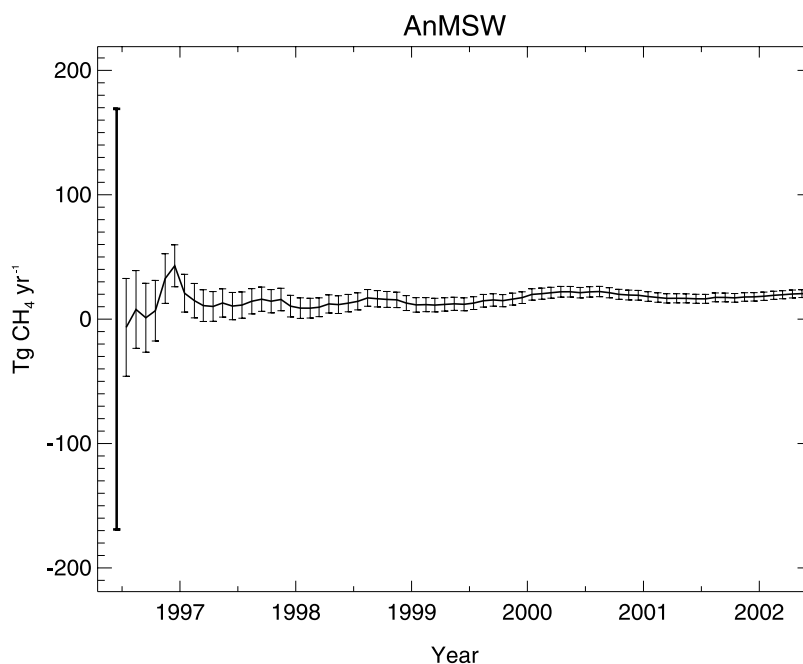


Figure 4. Kalman state vector evolution for the AnMSW methane source. The vertical axis corresponds to emission adjustments from the reference value ($\text{Tg CH}_4/\text{month}$), and the horizontal axis corresponds to the addition of new observational data. The initial vertical error bar corresponds to the a priori error associated with AnMSW. Unlike the seasonal sources the aseasonal sources are solved as constant fluxes over the entire 5-year time period. The optimized flux adjustment is obtained after all observations have been used.

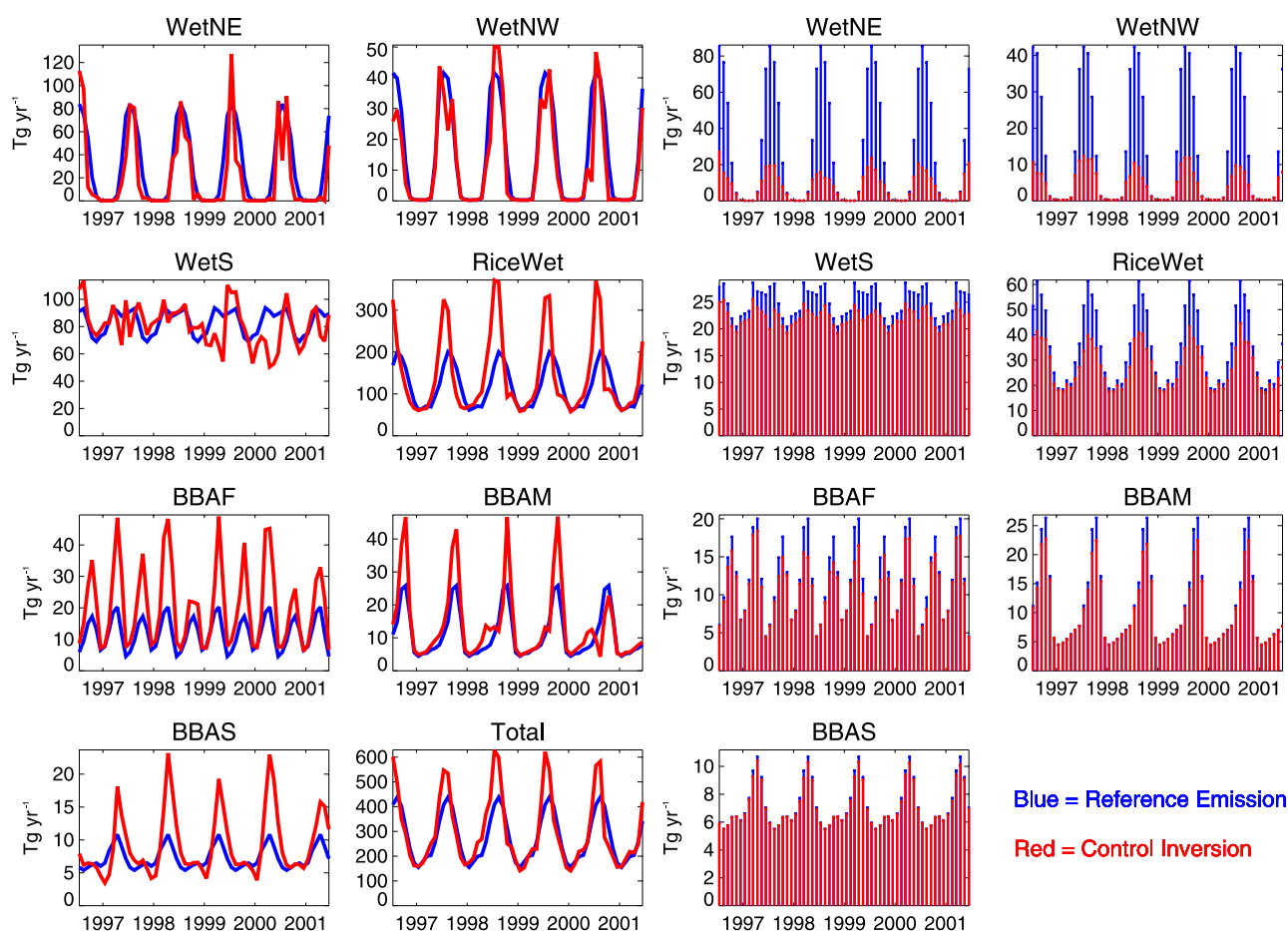


Figure 5. (left) Monthly optimized emissions in the control case and (right) corresponding uncertainties for the seven seasonally optimized sources. Note the different scales for the different sources. The reference emissions and uncertainties are shown in blue, and Control inversion case results are in red. Figure 5 (right) shows the error reduction from the reference (a priori) uncertainties.

improvements in model-observational comparison are apparent at nearly all sites, and used for interpretation of the inversion results in the following sections. As a further check of the inversion, we also verified that the monthly means in the optimized forward run and that produced by Equation 10 are in agreement.

5.5. Inversion Cases

[36] We investigated several different inversion cases that are based on different sets of actual and simulated observations as described in Table 2. The Control inversion uses 13 high-frequency (HF) and 41 flask sites as listed in Table 1, using monthly mean observations with MATCH output interpolated to exact observation times before averaging. The control case incorporates those 41 flask sites active for at least 70% of the 60 months between July 1996 and June 2001 (Table 1). The next case, HF only, uses only observations from the 13 high-frequency sites. This case is followed by HF MBL, which adds 21 flask sites located in the clean marine boundary layer (see Table 2). The next inversion, Control FullAv, uses the same observational sites as Control, but uses the full MATCH output to determine the modeled monthly means. As mentioned earlier, the model monthly means can be computed using modeled

mole fractions interpolated to the exact 4–5 flask sampling times each month (as in Control), or using all model time steps (FullAv cases) for that month.

[37] The Control Filt case also uses the same observational sites as Control, but uses filtered observations in which the pollution events have been removed for applicable sites (see Table 1). The Control FullAv Filt case combines the Control FullAv and Control Filt cases by using full MATCH monthly means and filtered observations. The next three cases, All Obs, All Obs Full, and All Obs Filt are analogous to Control, Control FullAv, and Control Filt, except that all 92 (rather than 54) available observation stations within the time period are used. The additional 38 observation sites have monthly data for less than 70% of the inversion months (Table 1). The distinction between sites with greater and less than 70% of the data is made because sites with significant amounts of missing data may affect the inversion results independently of their observational values. Most previous atmospheric inversions (e.g., Gurney *et al.* [2002] for CO₂) use a constant observational network, avoiding possible effects from the abrupt activity/inactivity of intermittent sites. These impacts are sometimes difficult to separate from emission changes due to the interannual variability of actual observations. The

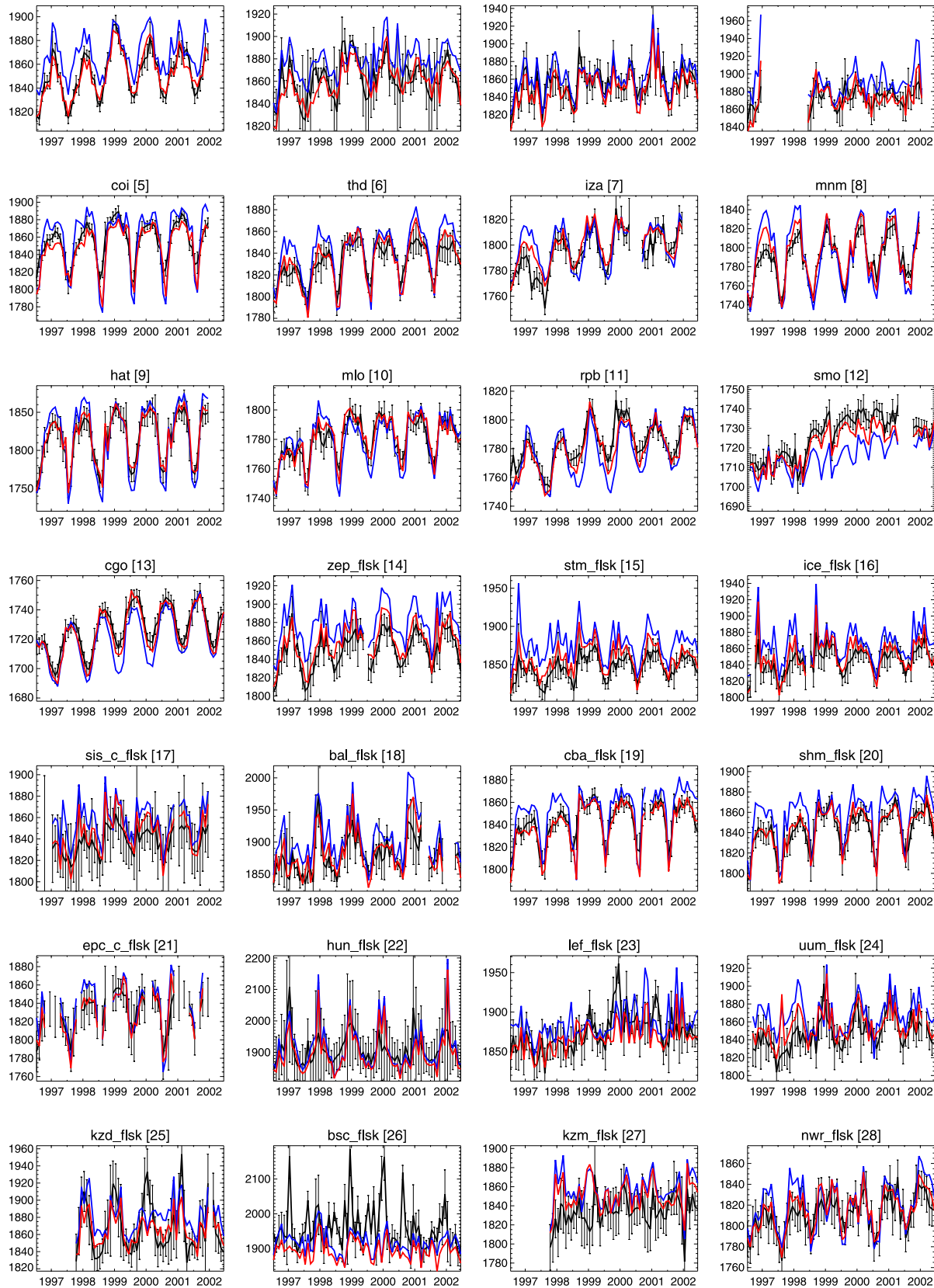


Figure 6. Monthly mean observations for all sites following the listing in Table 1. The observations (black) are shown with corresponding error bars described in section 5.1. The reference (blue) and optimized (red) curves represent MATCH monthly means (determined from model output interpolated to exact measurement time). The optimized values correspond to MATCH using the results of the All Obs inversion (see section 5.4).

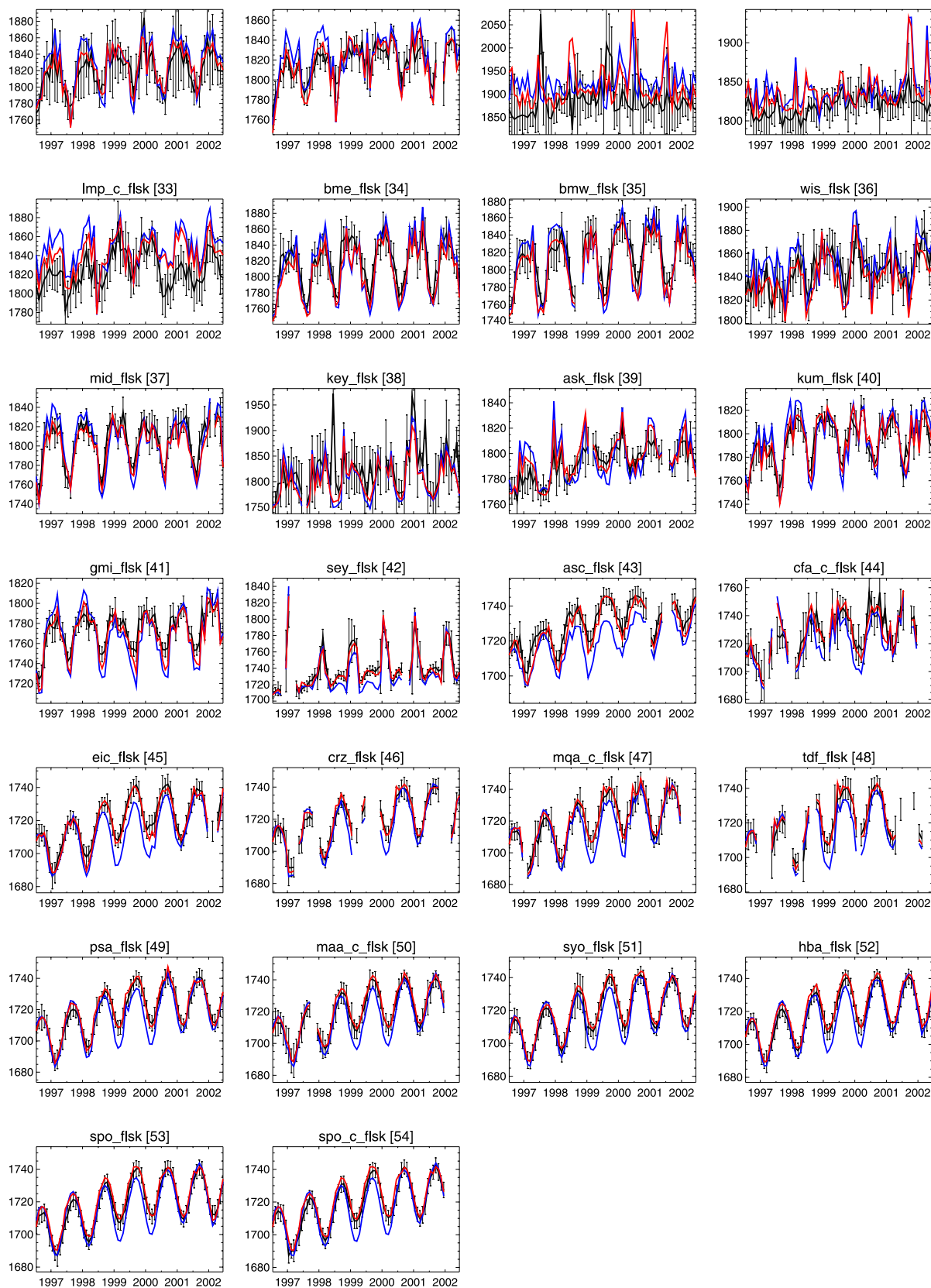


Figure 6. (continued)

70% cutoff is taken as an approximate value above which these impacts are minimized, but we also perform the inversion with all observations. Many of the inversion results for the 9 different cases are similar. We restrict the comparison across all cases to the annual average results (see section 6.2).

6. Average Inversion Results

[38] Although atmospheric CH₄ observations have been available since the early 1980s, only by the mid-1990s were sufficient stations operational to conduct an inversion using only high-frequency data at the regional scale. The inversion starts in July, 1996, one month before the 5th AGAGE station at Cape Matatula, Samoa began its high-frequency CH₄ measurements. The Kalman filter is formally run using nearly 6 years of observational data, until May, 2002. For the averaged inversion results described below, we consider the 5 year span between July 1996 and June 2001. This is because the last 11 months of the inversion includes monthly emissions which are suboptimized, as they do not incorporate a full 11 months of observations.

[39] Because of the large amount of methane emission information produced by the inversion, we divide the discussion of the methane inversion results into three temporal sets: average seasonal (i.e., annually repeating monthly fluxes), average annual (i.e., average fluxes over the entire inversion period), and interannually varying (i.e., monthly fluxes). This division also facilitates comparison to bottom-up estimates, which are often made at different temporal scales. The average seasonal cycles are derived by averaging the interannual monthly fluxes into a single annually repeating cycle for each source shown in Figure 5. To obtain the average annual results for the seven seasonal sources, we further average these monthly fluxes into annual mean fluxes applicable over the 5-year inversion period. The aseasonal sources are solved for as constant fluxes over the entire inversion period, and hence do not require averaging.

6.1. Average Seasonal Results

[40] The multiyear monthly averaged optimized emissions and emission uncertainties are shown in Figure 7. The flux values (Figure 7, left) are generated by averaging the monthly values between July 1996 and June 2001 of the interannually varying fluxes shown in Figure 5. Three inversion cases are shown for each source: Control (red), All Obs (orange), and HF (green). The original reference

case is shown in blue. The relationship $\bar{\sigma} = \sqrt{\sum_t^N \sigma_t^2 / N}$,

where $N = 5$ years and σ_t is uncertainty for a particular month (e.g., January), is used to compute the average uncertainties for each month (Figure 7, right).

[41] For the Control inversion, both WetNE and WetNW show decreases compared to the reference, particularly during the fall and spring. The overall decrease in the northern wetland regions is consistent with the reference run overestimate of the observed northern hemispheric mole fractions (Figure 6). These effects are particularly evident at the nearby stations of alt[1], brw[2], and frd[4] (Figure 6) where the reference overestimate has been corrected by the

inversion. WetS emissions show a dip during the spring months centered around May compared to the relatively smooth reference emissions, although emissions are still globally active throughout the year. The inversion strongly increases RiceWet emissions and shifts the maximum from August to July. The optimized emissions are decreased below the reference after September, partially offsetting the large increase in earlier months. At least part of the large increase can be attributed to the effects of measurement sites downwind of Asia, where the reference run exaggerates the seasonal mole fraction trough during June and July. This is most clearly seen in the reference mole fraction underestimates during the summer at coi[5] and hat[9] (Figure 6), which are two high-frequency stations most sensitive to RiceWet. The inversion compensates by shifting the RiceWet emissions to an earlier and more intense maximum, followed by a sharper decline. Note that an overestimate of tropical summertime OH in MATCH values could also lead to an overestimate of the reference seasonal mole fraction trough. However, the good representation of the methane seasonal cycle at Samoa, located in the remote tropics far from emissions, does not suggest a large error in the OH seasonality. The biomass burning (BBAM, BBAF, BBAS) regions have stronger emissions with greatly enhanced peak values relative to the reference. This result is consistent with the need for tropical emissions to exceed the reference values to better replicate the observations. The total emission for all seasonal sources (Figure 7) shows a significant increase above the reference. The shift of the seasonal source total to a July peak is mostly due to RiceWet, with smaller contributions from WetS and WetNE. Overall, the All Obs case is very similar to the Control, indicating that the 38 additional flask measurements listed in Table 1 do not significantly change the average seasonal results. For the more well-constrained regions such as WetNE and WetNW, the difference between Control and All Obs is indiscernible.

[42] Compared to the Control (and All Obs) inversion, the HF only case includes nearly offsetting increases and decreases in WetNE and WetNW, respectively. The presence of flask sites sensitive to Eurasian wetland emitting regions accounts for part of this difference. At flask sites that are sensitive to WetNE such as zep[14], stm[15], and bal[18], model simulations using the reference emissions overestimate observed CH₄. The decrease in WetNE emissions is smaller upon removal of these sites in case HF. The overall northern hemispheric decrease is shifted to WetNW, whose emissions are still sensitive to the high-frequency stations at alt[1], brw[2], and frd[4]. The WetS emission change is much smaller than the Control case, partially due to the fewer numbers of high-frequency stations to constrain this large and diffuse region. The strong increase in RiceWet is also produced, although to a slightly less extent than for the Control. The changes in the three biomass burning regions in HF also generally follow Control, although to a lesser degree. The HF seasonal total change is nearly identical to the Control, indicating that the global seasonal solution is largely independent of the addition of flask measurements to the high-frequency network.

[43] Figure 7 (right) shows the average monthly uncertainty reduction from the inversion. WetNE and WetNW have the greatest uncertainty reduction; these sources have

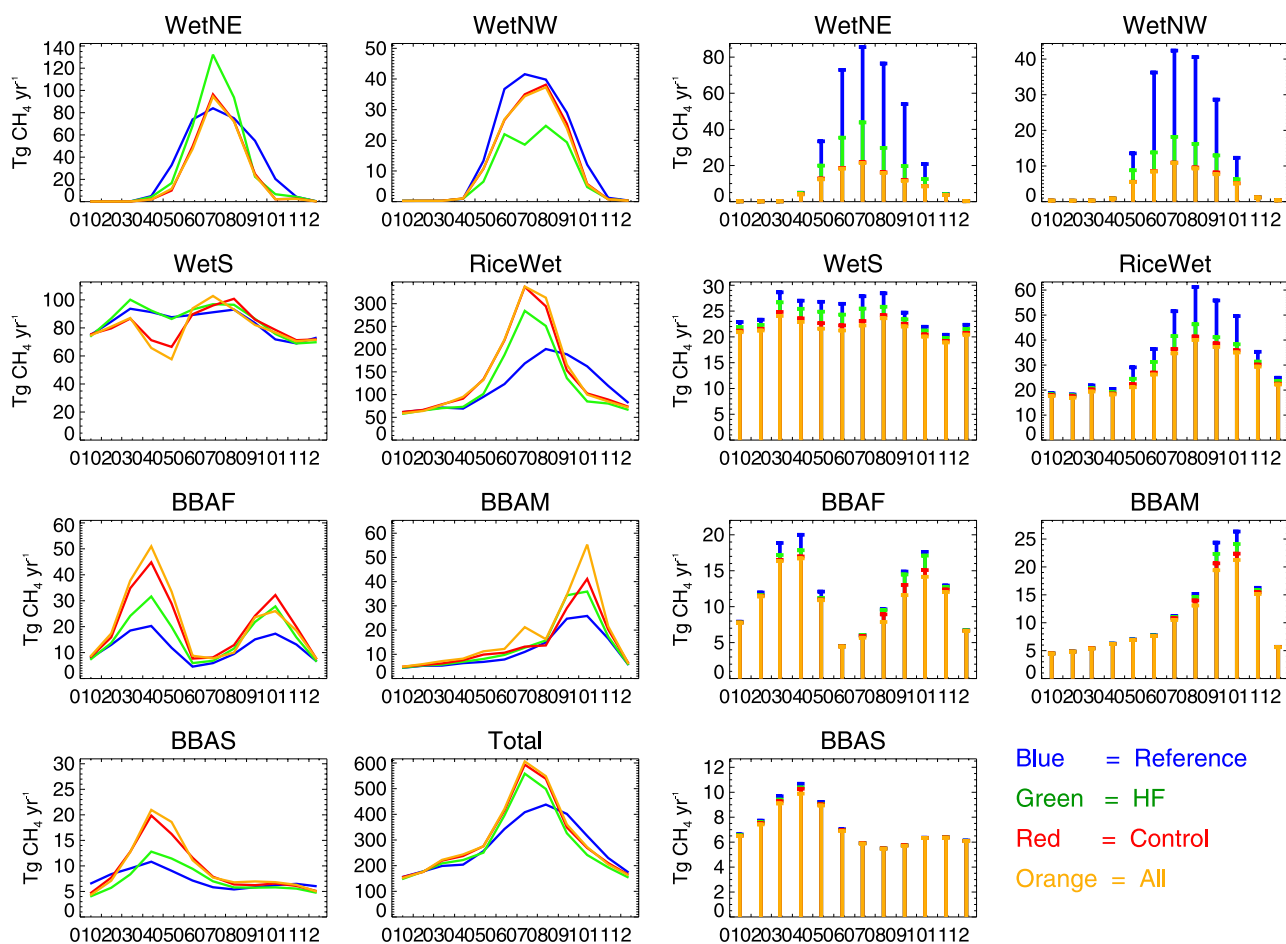


Figure 7. (left) Average seasonal cycle for optimized emissions and (right) corresponding uncertainties for the seven seasonally varying sources. Note the different scales for the different sources. The reference emissions and uncertainties are shown in blue, and inversion cases Control, HF, and All Obs are shown in red, green, and orange, respectively. Figure 7 (right) shows the error reduction from the reference (a priori) uncertainties.

several observational sites within and downwind of these regions. Moderate uncertainty reduction is seen for RiceWet, largely due to sites downwind of Asia. The least well constrained sources are WetS and the three biomass burning regions, which are all located in the tropics. The additional uncertainty reduction due to the flask network can be seen by the differences between the Control and HF error bars. The high-frequency sites account for most of the uncertainty reduction for WetNE, WetNW, and RiceWet, since there are sufficient nearby high-frequency sites to constrain these sources. The flask measurements provide additional uncertainty reduction for WetS, BBAF, BBAM, and BBAS. BBAS is the least well constrained even with flask measurements. Note that the additional 38 flask sites for All Obs lead to only small additional uncertainty reductions.

6.2. Average Annual Results

[44] The annually averaged optimized emissions (red) and emission uncertainties from the control inversions for aggregated processes are compared to reference emission (blue) in Figure 8. The numerical results for individual process emissions for all inversion cases were given in Table 3. The aggregated processes, including Wetland

(WetNE, WetNW, WetS, and part of RiceWet), and Biomass Burning (BBAM, BBAF, and BBAS) were listed in Table 2. The use of these aggregated emissions facilitates comparison to bottom-up estimates, as well as comparison between the nine inversion cases. The reference error bars in Figure 8 are the a priori uncertainties of each aggregated set of processes. The optimized case contains two sets of error bars. The right error bar corresponds to the spread in inversion values for all nine inversion cases in Table 2. The left error bar represents the uncertainty from the Control inversion. This uncertainty is always less than the reference uncertainty due to the influence of the measurements in the Kalman filter. The relationship $\bar{\sigma} = \sqrt{\sum_t \sigma_t^2 / N}$, where $N = 60$ months and σ_t is monthly error, is used to compute the annual average uncertainty for each seasonal process. The aseasonal uncertainties are taken from the last step of the Kalman filter as shown in Figure 4. That the uncertainties from the aseasonal components are much smaller than the seasonal components arises for two reasons. The first is that the aseasonal sources have very broad geographical

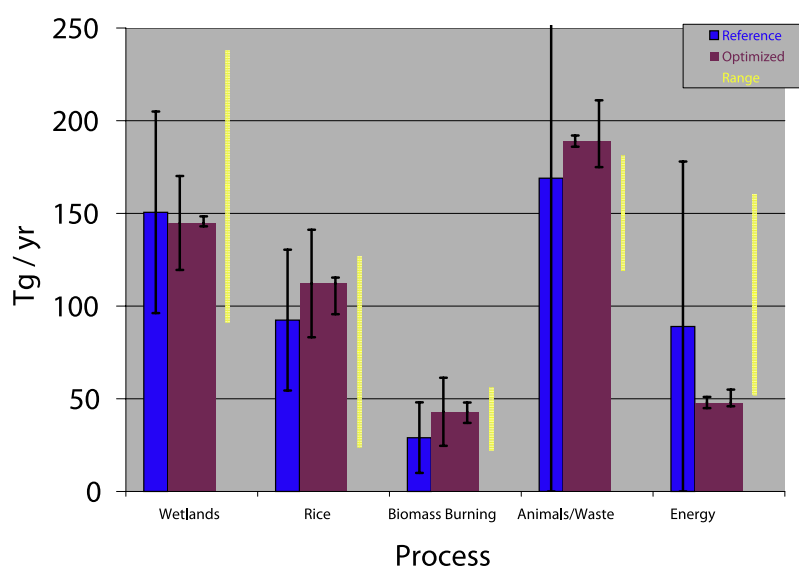


Figure 8. Annual average methane emissions. Shown are reference (blue) and optimized (red) emissions using the Control inversion. The error on the reference is the assumed a priori inversion uncertainty. The optimized values include two error bars: left bars corresponds to the inversion uncertainty for the Control case, and right bars to the spread of inversion results from the different inversion cases. The rightmost errors (yellow) represent the range of emission values found in the literature. See Table 2 for numerical values.

distributions and, consequently, strong sensitivities at many observing sites. The seasonal sources, in contrast, are more localized regionally and/or have relatively low sensitivities to the observing network (e.g., biomass burning). The second reason is that the inversion solves the aseasonal sources as constants over the entire time period, with flux uncertainty reduction for each new observation. The seasonal sources are solved as monthly fluxes which adds greater uncertainty to their 5-year averages.

[45] For the aseasonal processes, the inversion spread (right error bars) is larger than the inversion uncertainty (left error bars). For the seasonal processes, in contrast, the inversion uncertainties are much larger than the spread in inversion cases. This results from the seasonal processes being less well constrained than the aseasonal processes for the two reasons given above. The rightmost (yellow hanging) error bars in Figure 8 correspond to the range of estimates found in the literature for each process. The optimized values fall within this literature spread, although certain processes lie at the far end of that range. The total emissions are at the high end of the IPCC range of 500–600 Tg yr⁻¹, which is dependent on the prescribed OH sink, as discussed earlier. The higher total CH₄ emission contributes to the higher emission estimates for most processes when compared to the literature range.

[46] Compared to the reference case, the inversions increase AnMSW and decrease Energy emissions, respectively. As discussed previously, the inversion acts to decrease the reference run overestimate of the methane interhemispheric gradient in the reference run. An increase in AnMSW (globally distributed) and a decrease in Energy (largely northern hemispheric distribution) results in a net shift of emissions from northern to tropical and southern regions. Wetlands show some reduction, mostly from decreases in northern boreal emissions. RiceWet

emissions, which are dominated by sources between 0–30°N, increase by approximately 20% overall. Individual optimized tropical biomass burning sources are increased to between 30 and 80% above the reference. Table 2 indicates that the emission totals are similar among the nine optimized cases, since the total is determined largely by the prescribed global OH field. The addition of flask data generally decreases emissions from AnMSW and Wetlands, and increases Energy, Biomass Burning, and Rice emissions. Addition of flask data also leads to greater uncertainty reduction for each process. Note that compared to the Control case, the addition of 38 flask sites in All Obs leads to only slightly smaller uncertainty reductions, partially because these additional observations are active for less than 30% of the inversion period.

6.3. Comparison to Bottom-up Estimates

6.3.1. Wetlands

[47] *Cao et al.* [1996b] used a process model that incorporated substrate availability and climatic variables to estimate global wetland emissions of 92 Tg yr⁻¹. Their total is divided into northern, temperate, and tropical contributions of 23.3, 17.2, and 51.4 Tg yr⁻¹, respectively. Using another wetland process model, *Walter et al.* [2001b] estimated total wetland emissions of 260 Tg yr⁻¹, with a 25% contribution from wetlands north of 30°N. In both studies, most of the wetland emissions are located in the tropics. Our optimally estimated global wetland flux of 143–148 Tg yr⁻¹ (including a 21% contribution from RiceWet) lies between these two estimates but is closer to the *Cao et al.* [1996b] estimate. The dominance of tropical and southern emissions (~70%) over northern emissions (~30%) from the inversion is consistent with both process model studies, as is our estimated peak wetland emissions during July [*Cao et al.*, 1996b].

Table 3. Reference and Nine Inversion Results^a

Inversion Region	WetNE	WetNW	WetS	RiceWet	BBAF	BBAM	BBAS	AnMSW	Energy
Reference	29 ± 43	14 ± 22	83 ± 25	117 ± 38	11 ± 12	11 ± 13	7 ± 7	169 ± 169	89 ± 89
Control	22 ± 11	12 ± 6	81 ± 22	142 ± 29	20 ± 12	14 ± 12	9 ± 7	189 ± 3	48 ± 3
HF Only	29 ± 20	8 ± 9	86 ± 24	121 ± 31	16 ± 12	14 ± 13	7 ± 7	211 ± 4	46 ± 4
HF MBL	26 ± 17	10 ± 8	81 ± 23	139 ± 30	20 ± 12	14 ± 12	9 ± 7	184 ± 3	54 ± 3
Control FullAv	22 ± 12	12 ± 6	82 ± 22	141 ± 28	22 ± 12	14 ± 12	10 ± 7	180 ± 3	54 ± 3
Control Filt	22 ± 11	12 ± 6	81 ± 22	144 ± 29	21 ± 12	14 ± 12	9 ± 7	187 ± 3	48 ± 3
Control FullAv Filt	21 ± 12	12 ± 6	82 ± 22	145 ± 29	23 ± 12	15 ± 12	10 ± 7	175 ± 3	55 ± 3
All Obs	21 ± 11	12 ± 6	80 ± 22	143 ± 28	21 ± 11	17 ± 11	10 ± 7	185 ± 3	48 ± 3
All Obs FullAv	21 ± 11	12 ± 6	80 ± 22	146 ± 28	21 ± 11	17 ± 12	9 ± 7	182 ± 3	49 ± 3
All Obs Filt	21 ± 12	12 ± 6	82 ± 22	145 ± 28	21 ± 11	17 ± 12	10 ± 7	178 ± 3	52 ± 3

^aValues are in Tg CH₄ yr⁻¹. Section 5.5 describes the nine inversion cases.

6.3.2. Rice

[48] Over the past few decades, there has been a downward trend in the estimates of rice emissions using bottom-up methods [Mosier *et al.*, 1998]. This trend is largely due to the results from the increasing numbers of in situ flux measurements used for emission extrapolation, which collectively indicate lower overall rice emissions. Sass [1994] estimated global rice emissions between 25 and 54 Tg yr⁻¹ by combining rice cultivation area and flux estimates. Using a process based model of CH₄ emissions from rice, Cao *et al.* [1996a] estimated a global flux of 53 Tg yr⁻¹. The inverted emissions for the RiceWet emitting regions of 96–115 Tg yr⁻¹ (79% of RiceWet) are about twice current bottom-up rice estimates, but are closer to other top-down studies [Lelieveld *et al.*, 1998; Hein *et al.*, 1997]. This discrepancy may indicate a bias in the bottom-up extrapolations, but it may also partially arise from the presence of emissions within rice emitting regions that are included in our RiceWet emission region, but not accounted for in bottom-up studies. Bottom-up estimates focus on rice cultivation areas and may ignore emissions from nearby inundated regions formed by either natural or anthropogenic processes. Emissions during the nongrowing periods are also excluded in some estimates [e.g., Yan and Cai, 2003]. Incorporation of these other emissions would likely lead to larger bottom-up estimates from areas that include rice cultivation.

[49] In our inversion, the increase in rice emissions above the reference are most pronounced in July and August (Figure 7). The global seasonality of rice paddy emissions are difficult to estimate from bottom-up approaches because different regions have different rice planting seasons. For flooded rice fields, peak methane emissions usually occur several weeks after the initial flooding. The timing and magnitude of the fluxes depend strongly on soil characteristics, plant type, and fertilizer composition, in addition to climatological factors. Cao *et al.* [1996a] used a process model to estimate that over half of rice emissions occur between July and October globally, with peak emissions occurring in August. In a study of methane emissions from China, Yao *et al.* [1996] estimated peak methane emissions in June using regional classification and emission rates from six sites.

[50] Among the high-frequency sites, hat[9] and coi[5] are two sites most sensitive to rice emissions, and we note that these observations are reproduced well by the optimized MATCH run. The measurements at the most sensitive flask site, wlg[32] in north central China, are sometimes

overestimated by the optimized MATCH run during July and August. However, the seven nearby South China Sea observations (scsn) are not overestimated by MATCH (auxiliary material), although these stations are less sensitive to RiceWet emissions during the summer due to transport. This discrepancy among stations suggests that additional continental based measurements in this region should be made. The large observational error bars of wlg[32] (±23 ppb, Table 1) due to the sampling frequency error make future high-frequency measurements very desirable in this region.

6.3.3. Biomass Burning

[51] The optimization nearly doubles the reference biomass burning fluxes for all tropical regions, although the estimated emissions are still within the current literature range. Biomass burning emissions are poorly constrained in the inversion due to observational undersampling in the tropics. Bottom-up studies also have large uncertainties, which arise from the challenge of estimating total amounts of biomass burning. In addition, the associated CH₄ emission factor depends on the fire type and can be highly variable [Hao and Ward, 1993]. The only bottom-up value for global CH₄ emissions corresponds to our reference case from Hao and Ward [1993]. Our inversion confirms the bimodal behavior of BB Africa (Figure 7), but doubles its amplitude from the reference. This bimodal behavior is caused by seasonal shifts in the ITCZ, which lead to alternating dry and wet periods within a single region [Duncan *et al.*, 2003b]. Table 3 shows that total methane emission decrease from BB Africa, BB America, to BB Asia. This result is qualitatively consistent with the Hao and Ward [1993] estimate applicable for the 1980s. Duncan *et al.* [2003b] find greater CO emissions from Asia compared to the Americas during the 1990s, although the applicability of this result to CH₄ is unclear. More exact determination of the tropical biomass burning source for CH₄ will require more sensitive observations.

6.3.4. Aseasonal Processes

[52] The Intergovernmental Panel on Climate Change (IPCC) [2001] includes top-down and bottom-up estimates which range from 120 to 180 Tg yr⁻¹ for animal and waste (landfills, MSW, etc.) emissions [Fung *et al.*, 1991; Hein *et al.*, 1997; Houweling *et al.*, 1999; Lelieveld *et al.*, 1998; Olivier *et al.*, 1999]. Our optimally estimated total of 178–211 Tg yr⁻¹ for AnMSW is at the higher end of this IPCC estimate. The increases in AnMSW in our inversions are mostly offset by decreased Energy emission estimates (although this compensation is not a constraint in the

inversion). The IPCC total energy emissions range is 75–100 Tg yr⁻¹, based mostly on bottom-up studies using economic data. *Quay et al.* [1999] used global measurements of ¹⁴CH₄ to estimate that fossil (energy) methane emissions are 9–27% of the total methane emissions. This percentage range has been applied to the reference emission total of 590 Tg yr⁻¹ to compute the absolute range in literature values shown in Figure 8. The optimally estimated Energy emissions, which represent coal and natural gas emissions, total 46–55 Tg yr⁻¹. To compare to the total bottom-up energy estimates, an additional 18 Tg yr⁻¹ from various industrial sources (included in the reference run as constants but not solved by the inversion) are added, which leads to an adjusted energy range of 64–73 Tg yr⁻¹. Note that this optimized total is still at the lower end of the range of bottom-up energy literature estimates.

[53] Most of the literature estimates of energy emissions correspond to time periods before the 1996–2001 time period of our inversion. The EDGAR 3.0 data set shows decreases in the estimated gas and coal CH₄ emissions between 1990 and 1995. At least part of the decrease was associated with the collapse of the centrally planned economies of Eastern Europe and the former Soviet Union (FSU). Total gas production, and to a lesser extent coal, are reported to have decreased from these countries starting in the early 1990s [*Energy Information Administration (EIA)*, 2003]. *Dlugokencky et al.* [1994] also attribute observed atmospheric methane reductions to decreased gas leakage in FSU in the early 1990s. Coal production also decreased in China in the late 1990s due to the centrally planned closure of many small-scale mines and a switch to imported oil [*EIA*, 2003]. Coal emissions in the United States are also considered to have decreased by over 30% between 1990 and 2000 partially due to increased methane recovery efforts [*EIA*, 2002]. These reported energy trends may explain some of the difference between the older bottom-up literature values and the more current inversion results reported here.

6.4. Errors

[54] Although the observational mismatch error (see section 5.1) accounts for some aspects of model transport error, the inversion otherwise assumes a perfect atmospheric model. Systematic errors in transport, OH sink, and emission geographic patterns can contribute to biased inversion estimates. For example, deficiencies in the modeled interhemispheric transport and hence mole fraction gradient (IHG) can influence the annual average values. As mentioned in section 4, however, MATCH successfully reproduces the observed IHG of purely anthropogenic compounds such as SF₆ and CFC-11. The ability of the model to reproduce observations at many high-frequency CH₄ sites adds further confidence to its synoptic-scale transport. Errors in the spatial and temporal pattern of the OH field may also contribute to a bias error, but large errors are not detected from the methyl chloroform simulations over the inversion time period [*Chen and Prinn*, 2005]. The seasonal methane cycle at Samoa, dominated by tropical OH values and interhemispheric transport, and relatively insensitive to emission sources, is also well reproduced (Figure 6).

[55] Another possible source of error lies in the assumed emission spatial patterns used to solve for the emission magnitudes. This error is potentially greater for the large aseasonal sources that have many nearby and sensitive observing sites. A different assumed spatial emission pattern would alter the modeled sensitivity at these sites to these sources and, consequently, alter the inverted emissions. The sensitivity to the emission pattern is smaller for localized regions that do not contain observing sites, such as the tropical biomass burning regions. For these regions, a somewhat different assumed emission pattern would still likely result in a similar signal at a downwind site. The proper test of this error would be to use alternative emission patterns, especially for the aseasonal components. The scarcity of alternative spatial distributions, as well as the large computational burden associated with using even a single set of assumed flux spatial patterns, makes these tests difficult.

7. Interannual Results

[56] This section describes the monthly, interannual methane emissions of the seven seasonal regional sources. The interannual variability is illustrated by computing the monthly anomalies (deviations) of each year's seasonal cycles in Figure 5 from the optimized average seasonal cycles in Figure 7. The monthly anomalies for each emission source are shown in Figure 9 for the Control (blue) and HF (red) cases. For each source, the summation of anomalies over all months is zero. The global variability, computed as the summation of the individual sources, is also plotted. RiceWet and WetNE are the largest contributors to the interannual variability, followed by WetS and WetNW. The absolute contributions from the three biomass burning sources are small, although their variability relative to their emission magnitudes are comparable to the other processes. In addition to the actual amplitude of emission variability from a particular source, the sensitivity of the emission estimates to the observations also contributes to the amplitude of the inverted emission anomalies. For example, BBAS is poorly sampled by the observing network; this may contribute to its relatively low emission variability compared to WetNW despite having a similar total flux magnitude.

[57] Figure 9 shows that the Control and HF cases are generally similar, but have significant differences for certain months. For example, HF shows a much larger increase for WetNE in 1998 compared to the Control case. The anomalies for WetS and all three biomass burning regions are larger for the Control case, which incorporates many more flask observations sensitive to these regions. The global total variabilities, which depend on global changes in CH₄ observations, are similar for the two cases. Note that Figure 9 does not suggest a strong linear trend in the total seasonal methane emissions, but does indicate a significant emission increase in 1998, as next discussed in section 7.1. Figure 10 compares the interannual differences between the Control and FullAv cases. The month-to-month differences between the two cases are usually small, although differences for certain months are larger than the very small differences between their annual average values (Table 3). For WetNE and WetNW, the flux anomalies between the Control and

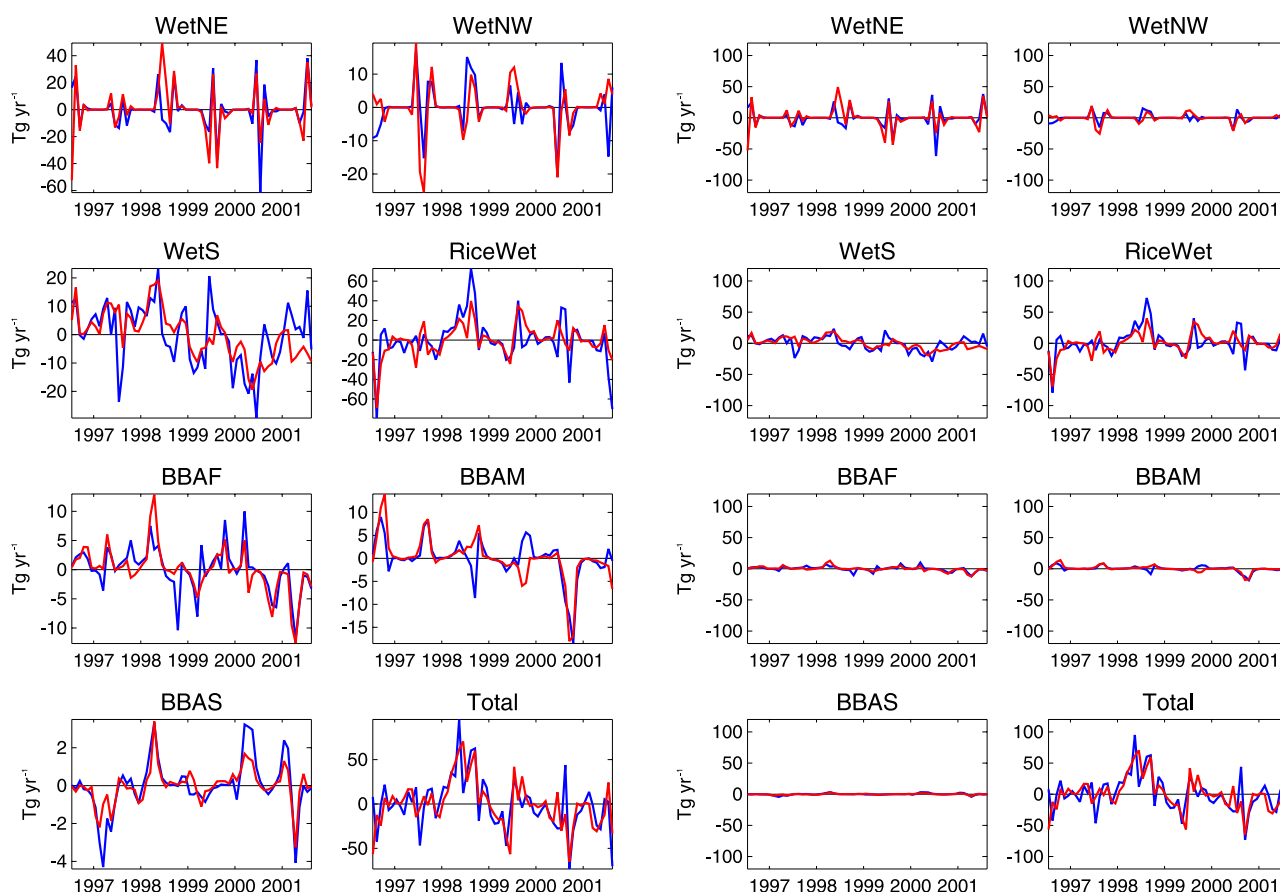


Figure 9. Monthly mean anomalies (from 5-year mean value) for Control (blue) and HF (red) inversion cases. Note the different vertical scales for different sources. Anomalies are repeated on the right side but with identical vertical scales to emphasize contributions of individual regional sources to the total change.

Control FullAv differ the most during the strongly emitting summer months. The Control anomaly is more negative than Control FullAv by about 10 Tg yr^{-1} for WetNE in May 1998. The Control FullAv peak anomaly is also centered in June, rather than July, 1998. Figure 10 shows the conflicting results that arise when using different sampling strategies are used to compute flask monthly means in the model and observations, and illustrates the additional uncertainty introduced when using flask observations in the inversion.

[58] We further compared the Control case and the All Obs case (Figure 11), showing the effect of adding the 38 additional flask stations which were active for less than the 70% of the months of the inversion (Table 1). Most of these additional observations affect the inverted values for the poorly constrained biomass burning emissions, rather than the already well-constrained emissions such as WetNE and WetNW. The shipboard measurements in the Pacific and the South China sea, which account for most of the additional observations in this 38-site database (Figure 6), are sensitive to BBAM and BBAS emissions, respectively. The new flask site in Namibia (nmb[63]) further constrains BBAF. These additional flask measurements increase the interannual variability (IAV) for biomass burning emissions. The peak heights for BBAF, for example, are greater in the All Obs case than the Control case (especially for BBAF). Note that the additional stations generally enhance the magnitudes of the monthly anomalies rather than alter their timing. This

suggests that although the smaller network can capture the general characteristics of the IAV, more sensitive observing sites are needed to accurately determine the full magnitude of emission anomalies. This argues for more long-term measuring sites directly downwind of these tropical regions. The total emissions are nearly the same between the Control and All Obs cases. We have also compared the Obs All and Obs All FullAv cases (not shown), which compares the effects of the different MATCH flask sampling strategies. The differences between the two cases are similar in magnitude to those between Control and FullAv (Figure 11.)

[59] Table 4 contain the flux anomalies averaged for each year for the Control, HF, and All Obs inversion cases. Total year-to-year emission changes fluctuate between $+33$ and -16 Tg yr^{-1} . This compares reasonably well with estimates by *Cunnold et al.* [2002], who used a semi-inverse method to compute global annual changes of up to $\pm 37 \text{ Tg yr}^{-1}$. The combined wetland values fluctuate between $+19$ and -15 Tg yr^{-1} . *Walter et al.* [2001b] used a wetland process model based on changes in precipitation and temperature to estimate annual changes of approximately $\pm 20 \text{ Tg yr}^{-1}$. Using the difference between Alert and Fraserdale mole fractions between 1990 and 1998, *Worthy et al.* [2000] estimated a wetland emission variability in the Hudson Bay Lowland (HBL), which is located within our WetNW region, between ± 0.23 to $\pm 0.5 \text{ Tg yr}^{-1}$. Assuming the HBL represents about 10% of northern

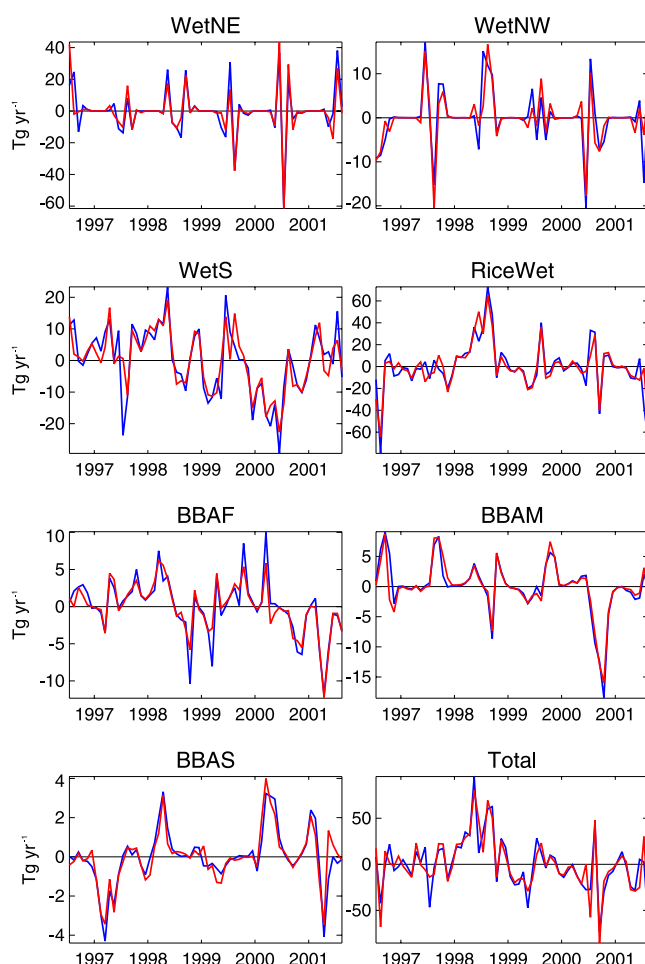


Figure 10. Monthly mean anomalies for Control and Control FullAv inversions. This plot compares the effects of using different flask sampling strategies.

hemispheric wetlands [Worthy *et al.*, 2000], this corresponds to a range of ± 2.3 to ± 5.0 Tg yr^{-1} for Northern wetlands, which is somewhat smaller than the ± 7 Tg yr^{-1} range for WetNE + WetNW emissions determined here. There have been fewer studies of the interannual variability in rice emissions. However, rice emissions can be expected to vary similarly to wetlands emissions, as both are influenced by climatic conditions (although the rice flooding stage is managed). Sass *et al.* [2002] measured emissions at a single site in the U.S over nine years and observed a year-to-year flux variability of approximately $\pm 50\%$ of the annual mean over the entire period. Biomass burning also fluctuates significantly from year to year [e.g., Duncan *et al.*, 2003b], although much less is known about associated methane emissions.

7.1. Global Methane Increase in 1998

[60] The ENSO event that occurred in late 1997 and 1998 influenced climate on a global scale [e.g., Bell *et al.*, 1999]. Global annual mean temperatures are considered to be highest for 1998 since the advent of reliable, direct measurements. In addition, both positive and negative precipitation anomalies led to regional flooding (e.g., central China in

June–July 1998, Indonesia in the second half of 1998) and drought (e.g., Indonesia in the first half of 1998), respectively. These climate changes likely contributed to the dramatic increase of global CH_4 mole fractions during this time [Dlugokencky *et al.*, 2001]. This large increase is clearly captured by the high-frequency observations at Samoa, shown in Figure 12. Figure 12 also shows the improvement in the model simulation when using Control optimized emissions compared to the reference. The interannual total methane flux anomaly in Figure 9 (right) shows a global CH_4 emission increase in 1998 followed by a steep decline during early 1999. Table 4 lists the optimized CH_4 anomalies between 1996 and 2001 for the Control inversion, which shows the unusually large total anomaly of $+33$ Tg yr^{-1} during 1998. This total anomaly falls between previously reported methane emission anomalies of $+24$ Tg yr^{-1} and $+37$ Tg yr^{-1} as estimated by Dlugokencky *et al.* [2001] and Cunnold *et al.* [2002], respectively, using hemispherically averaged mole fractions.

[61] Average surface temperature anomalies of approximately $+1^\circ\text{C}$ occurred over the northern and tropical land-masses [Bell *et al.*, 1999]. In general, increased wetland soil

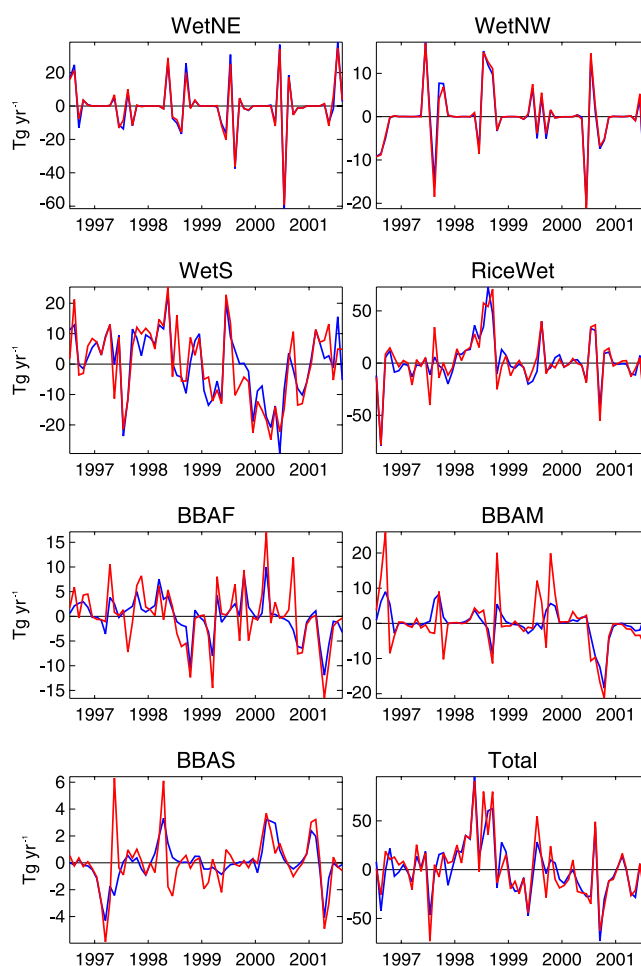


Figure 11. Monthly mean anomalies for Control and All Obs inversions. This plot shows the effect of using the extended flask stations.

Table 4. Inversion Annual Anomalies for Control, HF Only, and All Obs Cases^a

Region	1996			1997			1998			1999			2000			2001		
	Control	HF Only	All Obs	Control	HF Only	All Obs	Control	HF Only	All Obs	Control	HF Only	All Obs	Control	HF Only	All Obs	Control	HF Only	All Obs
WetNE	2.7	-2.6	2.8	-1.9	0.3	-1.3	1.5	8.9	1.6	-2.8	-6.7	-3.5	-2.0	-0.3	-2.3	2.5	0.5	2.8
WetNW	-1.9	0.3	-1.9	1.5	-1.2	0.7	2.2	-0.3	2.3	0.2	2.0	0.5	-1.6	-2.3	-1.4	-0.4	1.5	-0.3
WetS	2.5	2.4	2.6	3.2	5.2	3.0	5.3	8.3	7.4	-3.2	-3.5	-4.7	-10.7	-9.0	-11.9	2.8	-3.4	3.7
RiceWet	-7.5	-10.3	-5.6	-4.8	-4.5	-4.1	22.3	10.3	20.6	-1.4	4.5	-0.6	1.9	3.0	1.4	-10.4	-2.9	-11.7
BBAF	0.8	1.0	1.4	1.1	0.8	1.5	0.5	2.6	-1.1	0.5	0.0	-0.2	-0.6	-1.6	1.5	-2.3	-2.8	-3.1
BBAM	1.6	2.7	2.6	1.4	1.6	-0.7	0.5	1.9	1.3	0.5	-1.6	2.9	-3.7	-3.6	-4.4	-0.4	-1.0	-1.7
BBAS	-0.1	-0.1	0.0	-1.2	-0.8	-0.5	0.8	0.5	0.3	-0.3	0.0	-0.4	0.9	0.5	0.8	-0.2	-0.2	-0.2
Wet Total	1.7	-2.1	2.3	1.8	3.3	1.5	13.7	19.0	15.5	-6.1	-7.1	-7.9	-13.8	-10.9	-15.3	2.7	-2.1	3.7
Rice Total	-5.9	-8.2	-4.4	-3.8	-3.6	-3.2	17.6	8.1	16.3	-1.1	3.5	-0.5	1.5	2.4	1.1	-8.2	-2.3	-9.2
BB Total	2.3	3.6	4.0	1.3	1.7	0.3	1.8	4.9	0.5	0.8	-1.6	2.3	-3.4	-4.7	-2.1	-2.8	-3.9	-4.9
Total	-1.9	-6.7	1.9	-0.7	1.4	-1.3	33.1	32.0	32.3	-6.5	-5.2	-6.1	-15.7	-13.2	-16.3	-8.3	-8.3	-10.4

^aValues are in Tg yr⁻¹.

temperatures enhance methanogenic activity, leading to increased CH₄ production and emission. Anoxic environments are also necessary for methanogens to survive, and increased precipitation (expected during this El Niño year) generally increases anoxia in soils. Our inversely estimated emission anomalies are consistent with increased methane flux from wetlands in 1998. Rice emissions also show an increase in 1998 in our inversions; this result is not surprising because both natural and rice processes should have similar sensitivities to temperature and, to a lesser extent, precipitation. Table 4 contains the annual variability for the HF and All Obs inversion cases. Across all three inversion cases, the aggregated Wetlands and Rice 1998 anomaly flux range between 14 and 19 Tg yr⁻¹ and 8 and 18 Tg yr⁻¹, respectively.

[62] *Dlugokencky et al.* [2001] used the wetland CH₄ emission model of *Walter et al.* [2001a] to estimate 1998 flux anomalies. The model was driven by NCEP soil temperature and precipitation data from 1980 to 1999. The spatial distribution of the wetland model was further subdivided into northern and tropical wetlands based on the *Matthews et al.* [1987] distribution. This distribution is therefore spatially similar to our WetNE, WetNW, and WetS distributions, as well as the wetland contribution to the RiceWet distribution. In the bottom-up model, methane fluxes increased by approximately 20% for a 1°C temperature increase, and 8% for a 20% precipitation increase. The simulated 1998 CH₄ emission anomalies for northern and tropical wetlands were 12 and 13 Tg yr⁻¹, respectively, compared to the 1980–1999 average. These values are slightly larger than the northern and tropical wetland anomalies of 5–10 and 8–10 Tg yr⁻¹ for the three inversion cases in Table 4 (recall that 21% of RiceWet are attributed to wetlands). The inversion also produces a rice emission anomaly of 8–18 Tg yr⁻¹, which may include further overlapping wetland emissions, as discussed in section 3. *Mikaloff Fletcher et al.* [2004a], using an inverse approach which incorporated ¹³CH₄ observations, also suggest that wetlands dominated the 1998 observed methane increase. Unfortunately, no bottom-up estimates of rice emission anomalies for 1998 are available for comparison.

[63] Biomass burning is known to have increased for certain regions during 1998. With the exception of BBAF for the All Obs case, all biomass burning CH₄ emission anomalies are positive in 1998, although the magnitudes are smaller than for Wetland and Rice. Southeast Asia experi-

enced ENSO related biomass burning between August 1997 and April 1998. *Levine* [1999] estimated an anomalous Indonesian biomass burning emission of 1.9 Tg during the late fall of 1997. The 1998 Control emissions in Table 4 indicates an anomaly of 0.8 Tg yr⁻¹ from BB Asia, with most of the increase occurring in early 1998 as shown in Figure 9. The Obs All case shows a similar flux behavior (Figure 11), but with stronger peak and trough emissions. Unfortunately, many of these stations in Obs All, such as the ship tracks in the South China Sea and in the Pacific were not active during the 1998 El Niño event. The closest high-frequency site, Hateruma (hat[9]), does not indicate an obvious methane enhancement during the Southeast Asian/Indonesian fires of August–December 1997, probably due to its position upwind of these fire during these months. Central America also experienced large amounts of burning between April and June 1998 due to an ENSO related drought [*Duncan et al.*, 2003b]. The exact timing of this event is only weakly reproduced by the inversion results (Figure 9), which is likely due to the lack of nearby observing sites. For the three inversion cases described in Table 4, the overall range of 1998 from BBAM is between +0.5 to +1.9 Tg yr⁻¹. Using satellite data, *Duncan et al.* [2003b] and *van der Werf et al.* [2004] found smaller positive anomalies of biomass burning in Africa compared to Asia and America in 1998. The three inversion cases show a 1998 methane BBAF emission anomaly ranging between -1.1 to +2.6 Tg yr⁻¹. A small region in Eastern Siberia experienced biomass burning between July and September 1998 due to unusually warm and dry conditions. Using AVHRR data, *Kasischke and Bruhwiler* [2002] estimated methane releases of 2.9–4.7 Tg. Since these particular wildfire areas are not explicitly represented, the inversion would likely attribute this possible contribution to enhanced emissions of WetNE.

[64] Figure 12 shows that the optimized MATCH forward run compensates for about half the difference between the reference run and observations. Given the remote location of Samoa to strongly emitting regions, it is unlikely that any of the emissions (including aseasonal sources) can further compensate for this discrepancy without first adversely influencing stations more sensitive to these emissions. The mismatch between observed and optimized mole fractions at Samoa suggests that decreased OH concentrations may also have contributed to the increased 1998 methane growth rate. Using CH₃CCl₃ measurements, *Prinn et al.* [2001]

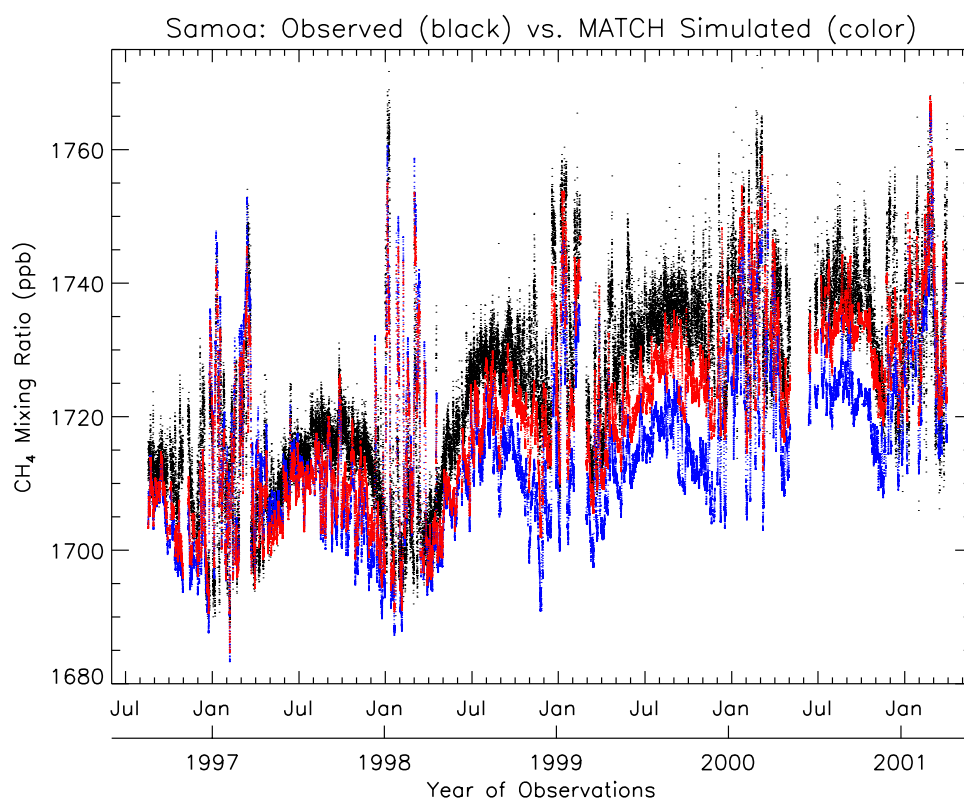


Figure 12. Observed (black) and simulated (red and blue) high-frequency mole fractions at Samoa. Red and blue correspond to MATCH simulations using reference and Control optimized emissions, respectively. Note the large increase in mole fraction observed in 1998. The optimized simulation (red) captures most, but not all, of this increase.

deduced anomalously low OH levels in 1997 and 1998. The driver for reduced OH concentration may be increased cloudiness (consistent with increased precipitation) which would reduce the amount of sunlight necessary for OH production. This effect would be pronounced for tropical regions where OH concentrations dominate CH_4 loss, such as Samoa. Novelli *et al.* [2003] linked the strong 1997–1998 increases in wildfires to large globally observed CO mole fractions; they further calculated significant decreases in global OH due to the large CO increase [see also Duncan *et al.*, 2003a].

7.2. Model Errors

[65] The model errors that affect the interannual results differ from those that affect the annual average results. Errors in the interhemispheric exchange rate and the model OH concentrations will bias the annual average modeled atmospheric CH_4 distribution, and consequently, the inverted annually averaged emissions. In contrast, the interannual emissions results depend less on average CH_4 distributions and more on observed and simulated monthly mole fraction anomalies. The deduced interannual emissions are thus less affected by large-scale transport biases which affect average CH_4 distributions, and more affected by errors in the short-term changes in the spatial flux patterns. For example, the inversion may not have been able to identify the boreal fires in Siberia during July–September 1998 because this pattern was not explicitly solved for. Instead, the inversion would have attributed this possible increase to WetNE emissions. The individual

seasonal patterns are also assumed to be representative for particular months, which becomes less true for very short time periods. A solution is to use even smaller process based emitting regions, ultimately at the grid resolution of the model, but the sparsity of observations make such an approach currently intractable at the global scale. A possible solution is to incorporate mesoscale models for regions sampled by nearby high-frequency observations [e.g., Kleiman and Prinn, 2000].

[66] The assumed constancy of aseasonal emissions is another simplification in this inversion. Although bottom-up studies indicate that the climate-driven seasonal processes considered here dominate the interannual variability, animals, waste, and energy emissions can also change year to year. Large fluctuations in these processes for the 1996–2001 time period have not been published. Possible changes in these aseasonal sources should, however, certainly be modeled for inversions over longer time periods. It has been hypothesized that reductions in gas and coal emissions led to the global decline of the CH_4 growth rate in the early 1990s [Dlugokencky *et al.*, 1994; Law and Nisbet, 1996]. As a sensitivity test, the inversion was also modified to solve for the linear aseasonal emission trends in addition to their average values. The inversion could not adequately constrain these small trends, because the trend sensitivities to the global measuring network were small and not sufficiently distinct. A trend of $+1 \text{ Tg yr}^{-1}$ added to each of the aseasonal source over the inversion period also had a negligible impact.

[67] Another potential model error involves using an annually repeating OH field, as discussed for the 1998 observed CH₄ mole fraction increase. Using methyl chloroform (MCF) data, *Prinn et al.* [2001] determined an increase followed by a decrease in the global average OH field between the late 1970s and the late 1990s. Incorporation of an optimized interannually varying OH field, derived from MCF studies for example, into the MATCH reference and sensitivity runs could then account for interannual changes in the sink.

8. Summary and Conclusions

[68] We have carried out an inverse modeling study of methane fluxes incorporating three new elements not previously combined in methane inversions: (1) high-frequency CH₄ observations, (2) interannual transport in the atmospheric transport model, and (3) the Kalman filter solution of interannually varying monthly fluxes. The first two elements are critical to determining methane emissions at higher space and time resolution. The development of new inversion techniques, such as done here using the Kalman filter, will also play a role in combining more realistic model and observational data.

[69] The annual average results of this study indicate a reduction in northern hemispheric emissions and an increase in tropical and southern hemispheric emissions compared to the reference. The deduced average seasonal maximum in emissions is shifted to July from the August reference, and is more intense. This result is dominated by phase changes and increases from rice emitting regions relative to the reference. The inverted emissions for wetlands compare well with other recent estimates. Inverted energy emissions are on the low side of prior estimates, but are consistent with expected recent decreases in methane emissions. The inversion produces rice emissions nearly twice the current bottom-up estimates. Part of the difference may be that the inversion solves for emissions from the entire rice emitting region, and includes those wetland emissions not formally associated with rice cultivation. Global wetlands and rice emissions dominate the observed CH₄ increase in 1998, consistent with a bottom-up study using a wetland processes model [*Dlugokencky et al.*, 2001]. The inverted methane flux increases for 1998 cannot fully reproduce the CH₄ growth rate at Samoa, suggesting that a decreased OH sink may also have contributed to the observed increase.

[70] On the basis of different sensitivity tests, the impact on the inversion results is influenced in order of decreasing influence by the following: observational network, modeled monthly means using exact observation time versus all model time steps, and filtered versus unfiltered observations. Several sources of model error are not easily input into the inversion framework. The use of alternative or additional emission patterns (such as for boreal biomass burning and soil uptake) can potentially affect the inversion results. As bottom-up studies continue to better define the spatial distributions of these fluxes, their inclusion into the top-down approach should result in more accurate flux estimates. This study also assumed an annually repeating OH sink. Future studies should incorporate interannually varying OH fields determined by other techniques, such as inverse studies of methyl chloroform. This study and *Chen*

and *Prinn* [2005] have shown the importance of larger-scale transport IAV. Another important aspect of model physics is the planetary boundary layer (PBL) transport for sites in strongly emitting regions. As more of these sensitive sites become active, tests of the accuracy of the PBL transport will become more crucial, perhaps using tracers that have better-known emission magnitudes compared to methane.

[71] The inversion leads to significant uncertainty reduction for northern wetlands, moderate reduction for rice, and small reductions for the tropical biomass burning and swamp sources. This clearly suggests that more methane observations are needed in the tropics to constrain biomass burning and wetland emissions from this region. The determination of the optimal placement of future observing sites (i.e., network design) have been conducted for CO₂ [*Patra and Maksyutov*, 2002; *Gloor et al.*, 2000]. The location of optimal observing sites depends not only the species of interest, but also the model to be used for the inversion. The most sensitive sites for emission estimates are often the most difficult to model, due to subgrid-scale effects. The network design algorithm should strike a balance between emission sensitivity, and model errors and resolution when determining optimal sites. An inversion algorithm similar to that used here could be used in principle to estimate the emission uncertainty reductions, and hence usefulness, of new observing sites.

[72] Methane isotopes (¹³CH₄, ¹⁴CH₄, CH₃D) may also aid in resolving isotopically distinct processes that have otherwise similar spatial and temporal patterns [*Mikaloff Fletcher et al.*, 2004a, 2004b]. The use of multiple tracers could aid in the attribution of methane sources, such as done for emissions off the Asian continent with other anthropogenic gases [*Bartlett et al.*, 2003]. Satellite data from sources such as MOPPITT and TES can provide global coverage of methane mole fractions, although information is often coarse in the vertical, and is less precise than ground based measurements.

[73] The use of a global, top-down approach to solve for specific interannual CH₄ flux regions/processes has reached the stage of allowing approximately monthly time resolution. Further increasing the time resolution of the optimization may require the optimization of spatially smaller emission regions, of which the smallest is ultimately the model grid resolution. The appropriate tool in this case would be an adjunct of the MATCH, which could in principle efficiently optimize the flux from each model grid cell. This technique has been applied to CH₄ [*Houweling et al.*, 1999] at coarse resolution. However, this technique would still require use of prior information in order to successfully constrain specific sources to avoid severe ill conditioning. Regional-scale modeling offers a complement to the global approach; for example, *Janssen et al.* [1999] and *Wang and Bentley* [2002] have estimated regional methane emissions using mesoscale models and high-frequency observations. Finally, future inverse studies may seek to estimate model parameters that link methane fluxes to climatological variables, as contained in CH₄ process models. This will lead to a true coupling between top-down and bottom-up approaches.

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References

- Bartlett, K. B., G. W. Sachse, T. Slate, C. Harward, and D. R. Blake (2003), Large-scale distribution of CH₄ in the western North Pacific: Sources and transport from the Asian continent, *J. Geophys. Res.*, **108**(D20), 8807, doi:10.1029/2002JD003076.
- Bell, G. D., M. S. Halpert, C. F. Ropelewski, V. E. Kousky, A. V. Douglas, R. C. Schnell, and M. E. Gelman (1999), Climate assessment for 1998, *Bull. Am. Meteorol. Soc.*, **80**(5), S1–S48.
- Cao, M. K., K. Gregson, S. Marshall, J. B. Dent, and O. W. Heal (1996a), Global methane emissions from rice paddies, *Chemosphere*, **33**(5), 879–897.
- Cao, M. K., S. Marshall, and K. Gregson (1996b), Global carbon exchange and methane emissions from natural wetlands: Application of a process-based model, *J. Geophys. Res.*, **101**(D9), 14,399–14,414.
- Chen, Y.-H. (2003), Estimation of methane and carbon dioxide surface fluxes using a 3-D global atmospheric chemical transport model, Ph.D. thesis, Mass. Inst. of Technol., Cambridge.
- Chen, Y.-H., and R. G. Prinn (2005), Atmospheric modeling of high- and low-frequency methane observations: Importance of interannually varying transport, *J. Geophys. Res.*, **110**, D10303, doi:10.1029/2004JD005542.
- Climate Monitoring and Diagnostics Laboratory (CMDL) (2001a), Methane, in *CMDL Summary Report 26*, Boulder, Colo.
- Climate Monitoring and Diagnostics Laboratory (CMDL) (2001b), Global-View-CH₄: Cooperative Atmospheric Data Integration Project—Methane [CD-ROM], report, NOAA, Boulder, Colo. (Available at ftp://ftp.cmdl.noaa.gov/ccg/ch4/GLOBALVIEW/)
- Conway, T. J., P. P. Tans, L. S. Waterman, and K. W. Thoning (1994), Evidence for interannual variability of the carbon cycle from the National Oceanic and Atmospheric Administration/Climate Monitoring and Diagnostics Laboratory Global Air Sampling Network, *J. Geophys. Res.*, **99**(D11), 22,831–22,855.
- Cunnold, D. M., et al. (2002), In situ measurements of atmospheric methane at GAGE/AGAGE sites during 1985–2000 and resulting source inferences, *J. Geophys. Res.*, **107**(D14), 4225, doi:10.1029/2001JD001226.
- Dentener, F., M. van Weele, M. Krol, S. Houweling, and P. van Velthoven (2003), Trends and inter-annual variability of methane emissions derived from 1979–1993 global CTM simulations, *Atmos. Chem. Phys.*, **3**, 73–88.
- Dlugokencky, E. J., K. A. Masarie, P. M. Lang, P. P. Tans, L. P. Steele, and E. G. Nisbet (1994), Reply to “Comments on ‘A dramatic decrease in the growth rate of atmospheric methane in the Northern Hemisphere during 1992’,” *Geophys. Res. Lett.*, **21**(22), 2447–2448.
- Dlugokencky, E. J., B. P. Walter, K. A. Masarie, P. M. Lang, and E. S. Kasischke (2001), Measurements of an anomalous global methane increase during 1998, *Geophys. Res. Lett.*, **28**(3), 499–502.
- Duncan, B. N., I. Bey, M. Chin, L. J. Mickley, T. D. Fairlie, R. V. Martin, and H. Matsueda (2003a), Indonesian wildfires of 1997: Impact on tropospheric chemistry, *J. Geophys. Res.*, **108**(D15), 4458, doi:10.1029/2002JD003195.
- Duncan, B. N., R. V. Martin, A. C. Staudt, R. Yevich, and J. A. Logan (2003b), Interannual and seasonal variability of biomass burning emissions constrained by satellite observations, *J. Geophys. Res.*, **108**(D2), 4100, doi:10.1029/2002JD002378.
- Energy Information Administration (EIA) (2002), Emissions of greenhouse gases in the United States, report, Dep. of Energy, Washington, D. C.
- Energy Information Administration (EIA) (2003), Country analysis briefs: Russia, report, Dep. of Energy, Washington, D. C.
- Enting, I. G. (2002), *Inverse Problems in Atmospheric Constituent Transport*, 392 pp., Cambridge Univ. Press, New York.
- Etiopie, G., and R. W. Klusman (2002), Geologic emissions of methane to the atmosphere, *Chemosphere*, **49**(8), 777–789.
- Fung, I., J. John, J. Lerner, E. Matthews, M. Prather, L. P. Steele, and P. J. Fraser (1991), Three-dimensional model synthesis of the global methane cycle, *J. Geophys. Res.*, **96**(D7), 13,033–13,065.
- Gloor, M., S. M. Fan, S. Pacala, and J. Sarmiento (2000), Optimal sampling of the atmosphere for purpose of inverse modeling: A model study, *Global Biogeochem. Cycles*, **14**, 407–428.
- Gurney, K. R., et al. (2002), Towards robust regional estimates of CO₂ sources and sinks using atmospheric transport models, *Nature*, **415**, 626–630.
- Hansen, J. E., and M. Sato (2001), Trends of measured climate forcing agents, *Proc. Natl. Acad. Sci. U.S.A.*, **98**(26), 14,778–14,783.
- Hao, W. M., and M. H. Liu (1994), Spatial and temporal distribution of tropical biomass burning, *Global Biogeochem. Cycles*, **8**, 495–503.
- Hao, W. M., and D. E. Ward (1993), Methane production from global biomass burning, *J. Geophys. Res.*, **98**(D11), 20,657–20,661.
- Hein, R., P. J. Crutzen, and M. Heimann (1997), An inverse modeling approach to investigate the global atmospheric methane cycle, *Biogeochemistry*, **11**(1), 43–76.
- Houweling, S., T. Kaminski, F. Dentener, J. Lelieveld, and M. Heimann (1999), Inverse modeling of methane sources and sinks using the adjoint of a global transport model, *J. Geophys. Res.*, **104**(D21), 26,137–26,160.
- Huang, J. (2000), Optimal determination of global tropospheric OH concentrations using multiple trace gases, Mass. Inst. of Technol., Cambridge.
- Intergovernmental Panel on Climate Change (IPCC) (2001), *Climate Change 2001: The Scientific Basis*, 944 pp., Cambridge Univ. Press, New York.
- Janssen, L. H. J. M., J. G. J. Olivier, and A. R. van Amstel (1999), Comparison of CH₄ emission inventory data and emission estimates from atmospheric transport models and concentration measurements, *Environ. Sci. Policy*, **2**(3), 295–314.
- Jockel, P. (2000), Cosmogenic ¹⁴C as tracer for atmospheric chemistry and transport, Ph.D. thesis, Ruprecht-Karls Univ. Heidelberg, Heidelberg, Germany. (Available at http://www.ub.uni-heidelberg.de/archiv/1426)
- Kasischke, E. S., and L. P. Bruhwiler (2002), Emissions of carbon dioxide, carbon monoxide, and methane from boreal forest fires in 1998, *J. Geophys. Res.*, **107**, 8146, doi:10.1029/2001JD000461. (Printed 108(D1), 2003.)
- Kleiman, G., and R. G. Prinn (2000), Measurement and deduction of emissions of trichloroethene, tetrachloroethene, and trichloromethane (chloroform) in the northeastern United States and southeastern Canada, *J. Geophys. Res.*, **105**(D23), 28,875–28,893.
- Law, K. S., and E. G. Nisbet (1996), Sensitivity of the CH₄ growth rate to changes in CH₄ emissions from natural gas and coal, *J. Geophys. Res.*, **101**(D9), 14,387–14,397.
- Lawrence, M. G., P. J. Crutzen, P. J. Rasch, B. E. Eaton, and N. M. Mahowald (1999), A model for studies of tropospheric photochemistry: Description, global distributions, and evaluation, *J. Geophys. Res.*, **104**(D21), 26,245–26,277.
- Lelieveld, J., P. J. Crutzen, and F. J. Dentener (1998), Changing concentration, lifetime and climate forcing of atmospheric methane, *Tellus, Ser. B*, **50**(2), 128–150.
- Lerner, J., E. Matthews, and I. Fung (1988), Methane emission from animals: A global high-resolution data base, *Global Biogeochem. Cycles*, **2**, 139–156.
- Levine, J. S. (1999), The 1997 fires in Kalimantan and Sumatra, Indonesia: Gaseous and particulate emissions, *Geophys. Res. Lett.*, **26**(7), 815–818.
- Mahowald, N. (1996), Development of a 3-dimensional chemical transport model based on observed winds and use in inverse modeling of the sources of CCl₃F, Ph.D. thesis, Mass. Inst. of Technol., Cambridge.
- Mahowald, N. M., R. G. Prinn, and P. J. Rasch (1997), Deducing CCl₃F emissions using an inverse method and chemical transport models with assimilated winds, *J. Geophys. Res.*, **102**(D23), 28,153–28,168.
- Matthews, E., and I. Fung (1987), Methane emissions from natural wetlands: Global distribution, area, and environmental characteristics of sources, *Global Biogeochem. Cycles*, **1**, 61–86.
- Mikaloff Fletcher, S. E., P. P. Tans, L. M. Bruhwiler, J. B. Miller, and M. Heimann (2004a), CH₄ sources estimated from atmospheric observations of CH₄ and its ¹³C/¹²C isotopic ratios: 1. Inverse modeling of source processes, *Global Biogeochem. Cycles*, **18**, GB4004, doi:10.1029/2004GB002223.
- Mikaloff Fletcher, S. E., P. P. Tans, L. M. Bruhwiler, J. B. Miller, and M. Heimann (2004b), CH₄ sources estimated from atmospheric observations of CH₄ and its ¹³C/¹²C isotopic ratios: 2. Inverse modeling of CH₄ fluxes from geographical regions, *Global Biogeochem. Cycles*, **18**, GB4005, doi:10.1029/2004GB002224.
- Milkov, A. V., R. Sassen, T. V. Apanasovich, and F. G. Dadshev (2003), Global gas flux from mud volcanoes: A significant source of fossil methane in the atmosphere and the ocean, *Geophys. Res. Lett.*, **30**(2), 1037, doi:10.1029/2002GL016358.
- Mosier, A. R., J. A. Delgado, and M. Keller (1998), Methane and nitrous oxide fluxes in an acid Oxisol in western Puerto Rico: Effects of tillage, liming and fertilization, *Soil Biol. Biochem.*, **30**(14), 2087–2098.

- Novelli, P. C., K. A. Masarie, P. M. Lang, B. D. Hall, R. C. Myers, and J. W. Elkins (2003), Reanalysis of tropospheric CO trends: Effects of the 1997–1998 wildfires, *J. Geophys. Res.*, **108**(D15), 4464, doi:10.1029/2002JD003031.
- Olivier, J. G. J., A. F. Bouwman, J. J. M. Berdowski, C. Veldt, J. P. J. Bloos, A. J. H. Visschedijk, C. W. M. van der Maas, and P. Y. J. Zandveld (1999), Sectoral emission inventories of greenhouse gases for 1990 on a per country basis as well as on 1[deg] × 1[deg], *Environ. Sci. Policy*, **2**(3), 241–263.
- Patra, P. K., and S. Maksyutov (2002), Incremental approach to the optimal network design for CO₂ surface source inversion, *Geophys. Res. Lett.*, **29**(10), 1459, doi:10.1029/2001GL013943.
- Prinn, R. (2000), Measurement equation for trace chemicals in fluids and solution of its inverse, in *Inverse Methods in Global Biogeochemical Cycles*, *Geophys. Monogr. Ser.*, vol. 114, edited by P. Kashibhatla, pp. 3–18, AGU, Washington, D. C.
- Prinn, R. G., et al. (2000), A history of chemically and radiatively important gases in air deduced from ALE/GAGE/AGAGE, *J. Geophys. Res.*, **105**(D14), 17,751–17,792.
- Prinn, R. G., et al. (2001), Evidence for substantial variations of atmospheric hydroxyl radicals in the past two decades, *Science*, **292**, 1882–1888.
- Quay, P. J., J. Stutsman, D. Wilbur, A. Snover, E. Dlugokencky, and T. Brown (1999), The isotopic composition of atmospheric methane, *Global Biogeochem. Cycles*, **13**, 445–461.
- Rasch, P. J., N. M. Mahowald, and B. E. Eaton (1997), Representations of transport, convection, and the hydrologic cycle in chemical transport models: Implications for the modeling of short-lived and soluble species, *J. Geophys. Res.*, **102**(D23), 28,127–28,138.
- Ridgwell, A. J., S. J. Marshall, and K. Gregson (1999), Consumption of atmospheric methane by soils: A process-based model, *Global Biogeochem. Cycles*, **13**, 59–70.
- Sass, R. L. (1994), Short summary chapter for methane, in *CH₄ and N₂O Global Emissions and Controls from Rice Fields and Other Agricultural and Industrial Sources*, edited by K. Minami, A. Mosier, and R. Sass, pp. 1–4, Natl. Inst. for Agro-Environ. Sci., Tsukuba, Japan.
- Sass, R. L., J. A. Andrews, A. J. Ding, and F. M. Fisher (2002), Spatial and temporal variability in methane emissions from rice paddies: Implications for assessing regional methane budgets, *Nutr. Cycl. Agroecosyst.*, **64**(1–2), 3–7.
- van der Werf, G. R., J. T. Randerson, G. J. Collatz, L. Gigio, P. S. Kasibhatla, A. F. Arellano Jr., S. C. Olsen, and E. S. Kasischke (2004), Continental-scale partitioning of fire emissions during the 1997 to 2001 El Nino/La Nina period, *Science*, **303**, 73–76.
- von Kuhlmann, R., M. G. Lawrence, and P. J. Crutzen (2003), A model for studies of tropospheric ozone and nonmethane hydrocarbons: Model description and ozone results, *J. Geophys. Res.*, **108**(D9), 4294, doi:10.1029/2002JD002893.
- Walter, B. P., M. Heimann, and E. Matthews (2001a), Modeling modern methane emissions from natural wetlands: 1. Model description and results, *J. Geophys. Res.*, **106**(D24), 34,189–34,206.
- Walter, B. P., M. Heimann, and E. Matthews (2001b), Modeling modern methane emissions from natural wetlands: 2. Interannual variations 1982–1993, *J. Geophys. Res.*, **106**(D24), 34,207–34,219.
- Wang, Y. P., and S. T. Bentley (2002), Development of a spatially explicit inventory of methane emissions from Australia and its verification using atmospheric concentration data, *Atmos. Environ.*, **36**(31), 4965–4975.
- Warwick, N. J., S. Bekki, K. S. Law, E. G. Nisbet, and J. A. Pyle (2002), The impact of meteorology on the interannual growth rate of atmospheric methane, *Geophys. Res. Lett.*, **29**(20), 1947, doi:10.1029/2002GL015282.
- Worthy, D. E. J., I. Levin, F. Hopper, M. K. Ernst, and N. B. A. Trivett (2000), Evidence for a link between climate and northern wetland methane emissions, *J. Geophys. Res.*, **105**(D3), 4031–4038.
- Wunsch, C. (1996), *The Ocean Circulation Inverse Problem*, Cambridge Univ. Press, Cambridge.
- Yan, X., and Z. Cai (2003), Methane emission from rice fields in mainland China: Amount and seasonal and spatial distribution, *J. Geophys. Res.*, **108**(D16), 4505, doi:10.1029/2002JD003182.
- Yao, H., Y. B. Zhuang, and Z. L. Chen (1996), Estimation of methane emission from rice paddies in mainland China, *Global Biogeochem. Cycles*, **10**, 641–649.

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